

A systematic approach to develop safety-related undesired event databases for Machine Learning analyses: Application to confined space incidents

Elena Stefana^{a,b,*}, Filippo Marciano^c, Nicola Paltrinieri^b, Paola Cocca^c

^a Department of Mechanical and Aerospace Engineering, Sapienza University of Rome, Italy

^b Department of Mechanical and Industrial Engineering, Norwegian University of Science and Technology, Norway

^c Department of Mechanical and Industrial Engineering, University of Brescia, Italy

ARTICLE INFO

Keywords:

Safety management
Risk prevention
Learning from incident
Safety-related decision making
Process safety
Learning task

ABSTRACT

In Occupational Safety and Health (OSH) and operational safety, confined spaces are high-risk working areas, where frequent serious and fatal incidents occur. However, there is a limited use of data-driven approaches based on Machine Learning (ML) techniques for learning from such incidents. In this context, our study proposes a systematic approach to develop a structured database suitable for ML-based analyses, considering various data sources with unstructured or semi-structured data. The approach was applied to 1346 incidents that happened in confined spaces mainly containing chemicals and located in manufacturing and process industries. This allowed building models predicting the fatal or non-fatal consequences and direct causes of incidents. The recognition of the approach potentialities and difficulties stimulated the development of an improved version of it to be adapted to different safety-related undesired events. The generalised systematic approach supports safety analysts during the process of learning from undesired events to maximise the lessons learned.

1. Introduction

In Occupational Safety and Health (OSH) and operational safety, a confined space is a space that: (1) is large enough and so configured that an employee can bodily enter and perform assigned work, and (2) has limited or restricted means for entry or exit, and (3) is not designed for continuous employee occupancy (OSHA, 2011). Silos, tanks, vessels, and sewers represent typical confined space examples, but there exist also unconventional ones, such as interior areas of machinery (Botti et al., 2022). It is not possible to provide an exhaustive list of confined space examples since some places may become confined spaces in certain situations or during the conducting of specific activities (HSE, 2013).

Confined spaces are high-risk working areas (Stefana et al., 2018), where the tolerance for errors or oversights is very small (McManus, 1998). McManus (2010) states that the term “confined space” is similar to a zipped computer file for pointing out the potential contemporary presence of multiple hazardous conditions and risks, including physical (e.g. caught in machinery, fall) and atmospheric (e.g. oxygen (O₂) deficiency or enrichment, toxic substances) hazards (McManus, 1998; Rekus, 1994; Selman et al., 2018; Stefana et al., 2021a). Furthermore, the conditions inside any confined space can be transient and change

rapidly from innocuous to life-threatening (McManus, 1998, 2010). These peculiarities caused frequent serious and fatal incidents in confined spaces (Chiu et al., 2020; Wang and Zhao, 2022). Different scientific articles highlight the worrying injury data in specific countries (e.g. Botti et al., 2018; Chiu et al., 2020; Ngah et al., 2022; Selman et al., 2018; Selman et al., 2019; Wang and Zhao, 2022).

To strengthen safety in confined spaces, safety regulations and standards have been specifically established over the years and worldwide: e.g. the US regulation No. 29 CFR 1910.146 (OSHA, 2011), the US ANSI/ASSP Z117.1–2022 (ANSI and ASSP, 2022), the Australian Standard AS 2865–2009 (SA, 2009), the UK Confined Spaces Regulations 1997 (Secretary of State, Department of the Environment, Transport and the Regions, 1997), the Italian DPR 177/2011 (DPR n. 177, 2011). Simultaneously, a relevant body of research has addressed the development of approaches and tools to be applied in the different steps of the risk management process to improve safety in confined spaces: e.g. a mobile application for the confined space identification (Botti et al., 2022), a five-step risk assessment tool (Burlet-Vienney et al., 2015a), possible applications of Industry 4.0 technologies (Stefana et al., 2018), solutions based on Inherently Safer Design (ISD) principles (Gonzalez-Cortes et al., 2022), and a procedure for a safe confined space entry rescue (Selman et al., 2019).

* Corresponding author at: Department of Mechanical and Aerospace Engineering, Sapienza University of Rome, Italy.

E-mail address: elena.stefana@uniroma1.it (E. Stefana).

Other scientific contributions perform investigations and conduct statistical analyses on confined space incidents, focusing on a specific type of consequence (e.g. Selman et al., 2018), a particular type of hazard (e.g. Chiu et al., 2020) or on a selected country (e.g. Xia et al., 2021). Data and reports about confined space incidents represent one of the main resources for exploring their root causes and contributory factors (Wang and Zhao, 2022). This is the key to “learning from incidents” (e.g. Liu et al., 2021) and “learning from the past” (e.g. Pasman, 2009), and is crucial to prevent the occurrence of similar events, limit damage, make progress, and improve safety (Lindberg et al., 2010; Pasman, 2009; Tamascelli et al., 2023). It is largely acknowledged that the usage of the lessons learned from past accidental scenarios represents a beneficial strategy to improve safety (e.g. Campari et al., 2023). Learning from incidents is also instrumental for modern safety management: any serious incident or accident provide a large learning potential (Simone et al., 2023).

There is a continuous need to make progress in learning from past incidents in an efficient way, not to lose and ignore the information recorded in incident data sources, and to render it available in a usable form to improve the safety level of organisations (Pasman, 2009). To guarantee these aspects, Machine Learning (ML) could represent a viable solution and incentive for learning from past lessons, enhancing the knowledge about hazardous scenarios, and detecting patterns and risk factors that human assessors may not be able to immediately recognise (Paltrinieri et al., 2019). ML techniques permit dealing with large dimensional data, extracting relations and rules among the attributes in data, predicting incident outcomes, injury risk and/or severity (Sarkar and Maiti, 2020). They can represent “the key to unlocking the value of safety data” to foster a proactive OSH and operational risk management (Nakhai et al., 2021). However, to the best of our knowledge, only one recent scientific article proposes the use of advanced data-driven approaches based on ML techniques for learning from confined space incidents. This article (Wang and Zhao, 2022) develops a framework to automatically extract 15 elements (e.g. date, type of confined space, fatality, injury, human error) and quantify the frequency of contributory factors from 325 incident records only written in Chinese and collected from a couple of data sources (safehoo.com and ichemsafe.com).

In such a context, our study investigates the potentialities of ML techniques for learning from confined space incidents, overcoming their current limited usage in the scientific literature. However, these incidents are collected from different data sources with unstructured or semi-structured data. That makes the adoption of data-driven methods not possible without a step of pre-processing of data and information. For this reason, this paper aims to propose a systematic approach to create a structured database (i.e. dataset) suitable for effective ML-based analyses of such incidents, by properly merging various and separate national and international data sources mainly described by means of text, unstructured or semi-structured data. Therefore, starting from incident records characterised by narrative descriptions and/or unstructured or semi-structured data available in more than one safety-related event sources, our approach permits: (1) obtaining a single dataset with all the relevant records, whose structure is derived from specific features of the undesired events under investigation, and (2) using such dataset as the key input to perform ML analyses and build models able to predict some features of interest about an event scenario. Note that this approach could also be applied to develop a single dataset for merging initial unstructured data sources with already structured data sources.

Such approach is primarily grounded on the confined space incidents, and then inductively generalised to be adapted to various kinds of undesired events in OSH and operational safety domains. This will allow enhancing both the processes of learning from confined space incidents and learning from safety-related undesired events.

The remainder of the paper is organised as follows. Section 2 summarises the methodology developed to build the structured database and perform ML analyses on confined space incidents. The obtained

database and results are presented in Section 3. Discussion about the process of learning from confined space incidents, its main potentialities and limitations, and a generalised version of the systematic approach are outlined in Section 4. Concluding remarks are provided in the final section.

2. Methodology

The methodology adopted for building a structured database about confined space incidents and performing ML analyses is depicted in Fig. 1. Such methodology enhances the strategy for the database development presented in Stefana et al. (2022a) by means of feedback and iterations among the steps to refine the intermediate obtained outcomes. Throughout the methodology, in accordance with Goasdoué et al. (2007) and Hoxmeier (1998), we addressed some database quality issues, such as integrity, quality, and completeness.

2.1. Define the purpose of the analysis and undesired events

The first step consists in defining the purpose of the analysis and the undesired events of interest. The purpose of our analysis was to learn from confined space incidents that are collected from various and separate national and international data sources. The undesired events under investigation were incidents occurring in confined spaces in different countries and leading to different kinds of consequences (i.e. fatalities and severe injuries) to one or more workers. In particular, we focused our attention on specific types of confined spaces: bin, cistern, container, dumpster, hopper, hopper car, silo, tank, tank car, tanker, vat, and vessel. They represent classical and cross-sector examples of confined spaces, which are not intended for human occupancy, but used for storing and/or carrying materials and substances, usually with limited access, and where equipment and mechanical devices may be present. Such combination of characteristics makes them the most hazardous kind of confined spaces (NIOSH, 1979), as emphasised by both Italian and international (older and more recent) data.

During the step of undesired event definition, we also established a set of inclusion and exclusion criteria, which are summarised in Table 1 (also reporting some examples of excluded records). These criteria were specified in an initial meeting among the authors and refined during a subsequent discussion to assure a certain level of accuracy and consistency of our database.

2.2. Characterise available data sources

Since the incident records are collected from various and separate national and international data sources, the main and most updated ones containing confined space injuries and fatalities should be taken into account. Also considering the databases mentioned by Botti et al. (2018), we consulted and selected the US Chemical Safety Board (CSB), the Italian National Institute for Insurance against Accidents at Work (INAIL), the US National Institute for Occupational Safety & Health (NIOSH), and the US Occupational Safety and Health Administration (OSHA) websites and archives. Among them, only NIOSH provides a list specifically on confined space incidents. In OSHA and INAIL websites, we searched for both fatality and severe injury investigations that are available in separate sources. Each data source was queried by means of the keywords indicating the specific confined spaces of our interest (i.e. bin, cistern, container, dumpster, hopper, hopper car, silo, tank, tank car, tanker, vat, and vessel), from the date of the oldest record in order not to exclude potentially relevant events.

2.3. Collect and select records

We collected the records derived from the selected data sources into six different databases reporting the incident descriptions (written in English or Italian) and other data and information present in the original

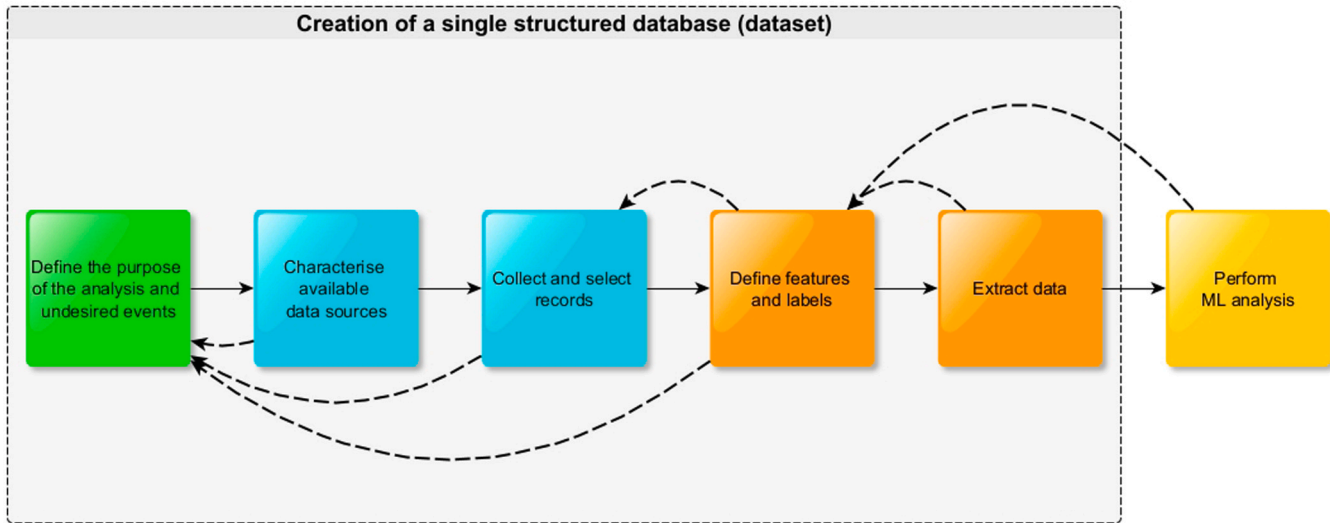


Fig. 1. Systematic approach for database development and ML analyses.

Table 1
Inclusion and exclusion criteria.

Inclusion criteria	Exclusion criteria
Compliance of the confined space definition provided by (OSHA, 2011)	Description of a near miss (e.g. OSHA 202495966)
Presence of one or more keywords related to the investigated confined spaces	Description of a suicide (e.g. OSHA 706556)
Consideration of the confined space as a significant factor for the workers' consequences, similarly to Selman et al. (2018) (e.g. one operator performing an activity outside the confined space and then fell into it inadvertently)	Presence of one or more investigated keywords not indicating confined spaces, such as ballast tank and M-1 tank (e.g. OSHA 2015107761)
Involvement of work personnel	Description of an incident related to a vessel indicating a boat or ship (e.g. OSHA 200556629)
Performance of work activities	Lack of details about the direct cause of the incident and of the workers' consequences (e.g. NIOSH report No. 1985–26)
	Presence of an undetermined or unspecified cause (e.g. OSHA 201163045)
	Lack of the identification of a precise direct cause of the incident and of the workers' consequences among a set of assumed conflicting causes (e.g. OSHA 14414627)
	Mentioning of a generic harmful substance, without clarifying its type, name, or effects on human body (e.g. INAIL code 1936)
	Presence of conflicting pieces of information (e.g. OSHA 200554160)

sources. We analysed and reviewed manually the retrieved records to select only those of interest: each author independently examined a subset of records in accordance with the inclusion and exclusion criteria defined previously. Therefore, a record was included if it met the inclusion criteria, otherwise it was excluded from the next step of our study. Proper brainstorming sessions were organised among all the authors to discuss any doubts and/or uncertainties about the inclusion / exclusion of some records. This permitted increasing the confidence level about the reliability of our results.

At the end of the step related to the record analysis and selection, we performed cleaning operations on each database: (1) removal of the duplicates (i.e. similar records corresponding to the same entity, as

reported by Goasdoué et al., 2007), (2) elimination of some columns not useful for the analysis and whose data were present in only one data source (e.g. national codes indicating the injury nature) or in few records of a data source (e.g. age of the involved worker), and (3) combination of different inspections and/or summaries of the same incident. These operations permitted merging the six analysed databases in a unique and initial database (in.xlsx format), where each row identifies an incident.

A further step of data cleaning was carried out to recognise any other duplicates existing among the different databases. This data deduplication step was easily implemented by filtering and sorting the records by incident dates, inspection codes, and/or record identifiers.

2.4. Define features and labels

To both describe unambiguously and in the same mode the undesired events whose data and information are arranged differently in each data source, the records in a database should be represented using the same set of features and corresponding labels (Kotsiantis et al., 2006). A feature is a “measurable property of an object or event with respect to a set of characteristics”, and provides a machine-readable way to describe the relevant objects (ISO and IEC, 2022). The classes of the object under investigation are defined as “labels” (or categories) (Tamascelli et al., 2020).

The features and labels should be defined in a way that all the relevant pieces of information and data in the reports are structured in dimensions able to:

- maintain an identifier of the data source collecting the incident;
- contextualise the incident in time and space;
- describe the scenario from the OSH and operational safety perspectives;
- take into account all the phases of the risk management process (i.e. hazard identification, risk estimation, risk evaluation, and risk treatment, in accordance with ISO and IEC (2014) and ISO (2018a, 2018b));
- capture the peculiarities of the confined spaces (also considering any previous studies about statistical incident analyses and contributory factors).

For such purpose, we defined the features and corresponding labels for properly describing the confined space incidents under investigation. Further analysis of confined space literature (including statistical incident analyses and contributory factors) and various consultations among the authors allowed identifying a set of desired features that a

database should have.

The identified desired features were evaluated against the information and data actually available in the majority of the investigated databases: this represented the step of feature selection and reduction in our approach that allowed us not to introduce too missing values and to control the overall database completeness and relevancy. The feature selection was then followed by the label definition step, which was performed also considering the literature about confined spaces.

2.5. Extract data

Differently from Wang and Zhao (2022), we did not employ any Natural Language Processing (NLP) methods to extract data from the records. NLP regards the discipline exploring how computers can be used to understand and manipulate natural language text or speech to perform a particular task (Chowdhury, 2003). NLP was not used for this research since: (1) the data sources are written in different languages (English and Italian), (2) the presence of words that are fragmented or misspelled (mainly, in OSHA data source), (3) the structure of the incident descriptions among the considered databases are markedly different, (4) the words and synonyms used to express the same concept are various among the databases (this is also related to the presence of different standards or regulations in the time period covered by our research).

Such peculiarities (that also represent some challenging tasks in NLP highlighted, for instance, by Khurana et al., 2023) could severely limit the opportunities offered by NLP and decrease its performance. Therefore, we manually assigned the label to each feature and for each record by reading the description of the incidents and other data available in the original databases (e.g. inspection pages available in the OSHA website). When the incident reports detail certain features for different workers, we assigned the labels considering the most severe injury and/or the most injured worker.

During the step of label assignment, we realised that some records could have been better described by other, previously undefined, labels. This gave the opportunity to refine and consolidate the final sets of labels populating our database. Finally, we completed the database replacing the missing values in the cells with the symbol “-” since there is no way to guess the lacking values through statistical estimations (Tamascelli et al., 2020). We therefore obtained the final structured database.

2.6. Perform ML analysis

The obtained dataset represents the key input for applying ML techniques. ML techniques can perform supervised, unsupervised, or reinforcement learning (Kotsiantis, 2007). Numerous ML applications involve tasks that can be set up as supervised (Kotsiantis et al., 2006), and this is the focus of our research. Supervised learning is carried out when the records (i.e. instances) in the dataset are given with the corresponding correct outputs (i.e. known labels) (Kotsiantis et al., 2006) with the purpose to predict future outcomes for new input instances (Sen et al., 2020). Supervised learning problems can be regression and classification ones: the former deals with continuous outputs, while the latter with categorical outcomes (Sen et al., 2020).

Among the different possible learning tasks that can be investigated, we selected the prediction of the fatal or non-fatal consequences of incidents and the direct cause of these consequences. For such purposes, classification models (i.e. classifiers) appear relevant. A classifier is a ML model whose expected output for a given input is one or more class labels (ISO and IEC, 2022). Therefore, we performed ML analyses by focusing on supervised classification models and, in particular, we employed Logistic Regression (LR), Random Forest (RF), and k-Nearest Neighbours (kNN) as classification ML algorithms. We selected these three algorithms since they represent state-of-the-art ML techniques and are very popular among the safety science researchers and practitioners

(e.g. Sarkar and Maiti, 2020). Furthermore:

- LR is one of the most important statistical and data mining techniques, is a standard method for building models for classification, produces class-probability estimates, can also extend to multi-class classification problems, performs better for smaller data sets, and is preferable where the classes cannot be separated well (Maalouf, 2011; Perlich et al., 2003);
- RF is an ensemble learning algorithm that consists of a collection of tree-structured classifiers, is an effective tool in prediction, and is characterised by an enhanced predictive power in comparison to a single decision tree (Breiman, 2001; Sarkar and Maiti, 2020);
- kNN is one of the top ten algorithms in data mining, is an easy to understand and implement classification technique, can perform well in many situations, is particularly well suited for applications in which an object can have many class labels, works well with large training dataset, but it requires high computational time consumption (Sen et al., 2020; Wu et al., 2008).

To implement such techniques, we used Orange data mining software (Orange). It is a ML and data mining suite for data analysis, which can be used by both experienced users and non-experts (Demšar et al., 2013). Orange permits applying ML techniques by building a workflow, which is a combination of widgets, in Orange’s visual programming environment. The workflow developed for our study is depicted in Fig. 2, and is composed of the widgets summarised in Table 2.

Orange permits obtaining different traditional performance metrics to evaluate the predictions (as reported in Sammut and Webb, 2017, and Witten et al., 2017), including:

- True Positive (TP) and True Negative (TN), i.e. the correct classifications;
- False Positive (FP), i.e. the outcome is incorrectly predicted as positive when it is actually negative;
- False Negative (FN), i.e. the outcome is incorrectly predicted as negative when it is actually positive;
- Area Under the Curve (AUC), i.e. the area under the Receiver Operating Curve (ROC), whose range varies from 0 to 1, where 0 means that every negative is scored higher than every positive, and 1 means that the classifier scores every positive higher than every negative;
- Classification Accuracy (CA), i.e. the ratio between the number of correctly classified objects and the total number of objects;
- F1, i.e. the harmonic mean of precision and recall (for two-class classification problems);
- Precision, i.e. the ratio of TPs and the total number of positives predicted by the model;
- Recall, i.e. the ratio of TPs and the total number of actual positives.

3. Confined spaces: incident database and ML analysis

3.1. Incident database

The approach described in Section 2 permitted developing a single structured database of fatal and non-fatal incidents that occurred in particularly hazardous types of confined spaces, i.e. bin, cistern, container, dumpster, hopper, hopper car, silo, tank, tank car, tanker, vat, and vessel. Such database was obtained by merging unstructured or semi-structured event data contained in the CSB, INAIL, NIOSH, and OSHA data sources and arranged differently among these sources. In the database, each row represents an incident, while each column a feature of the incident. A set of 48 features was defined and selected: the complete list of the features and their corresponding labels are reported in Table A1 in Appendix. For certain features, the complete group of labels is composed of both single and combined ones: combined labels are indicated by listing two or more single labels separated by the symbol “-”

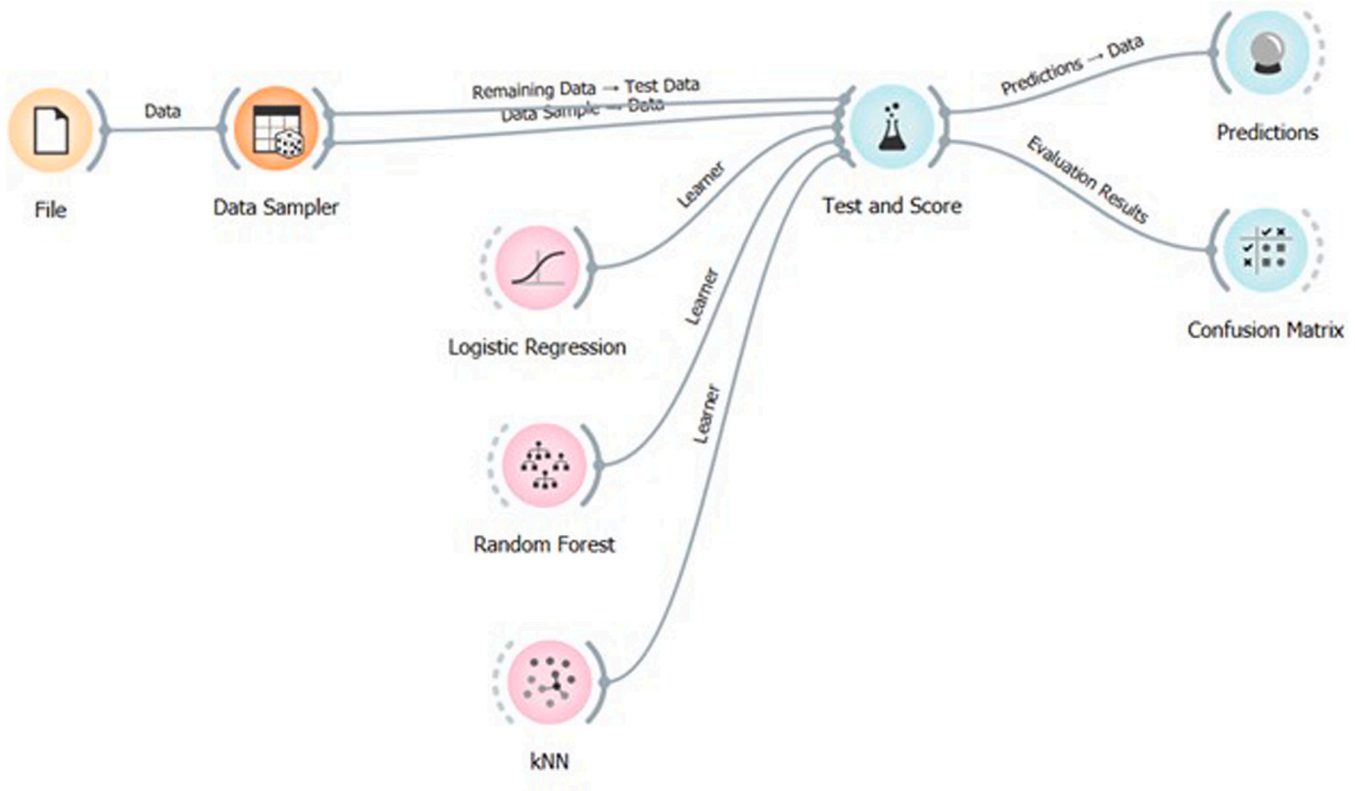


Fig. 2. Workflow created in Orange for the ML application.

Table 2
Widgets, their functions, and inputs of the workflow.

Widget	Function	Objective	Inputs and/or parameters
File	Data	To read an input data file in a suitable format and specify the type and role of each feature	Database; Target feature
Data Sampler	Transform	To select a subset of data instances from the input dataset for obtaining Data Sample (considered as training data) and Remaining Data (considered as test data)	Fixed proportion of data (70% of all the data) as the sampling method
Logistic Regression	Model	To use the LR learning algorithm	Regularisation type: Ridge (L2); Strength cost equal to 1
Random Forest	Model	To use the RF learning algorithm	Number of trees equal to 10; Not splitting subsets smaller than 5
kNN	Model	To use the kNN algorithm for making predictions	Number of neighbours equal to 5; Metric: Euclidean; Weight: Uniform
Test and Score	Evaluate	To test the learning algorithms on data and obtain performance measures	Test on test data
Predictions	Evaluate	To display the data and predictions produced by the models on the data	-
Confusion Matrix	Evaluate	To provide the number or proportion of instances between the predicted and actual class	-

or by means of the term “multiple”. Table A1 also reports the percentage of completeness for each feature, as suggested by Goasdoué et al. (2007).

Each feature and the corresponding labels can be displayed and analysed individually or in combination with others by means of some interactive and dynamic graphs and dashboards. Fig. 3, Fig. 4, Fig. 5, and Fig. 6 represent some examples of such graphs and dashboards that summarise some interesting properties of the created database.

3.1.1. Incident contextualisation and industry characteristics

As can be deduced from Fig. 3(a), the database collects 1346 confined space incidents, which are mainly derived from the OSHA data source (1197 records - 88.93%). Such events happened in US and Italy, between 1984 and 2021, covering a period of 38 years (Fig. 3(b)). The temporal distribution also highlights that in the years 1990, 1994, and 2017, more than 50 annual incidents took place. Manufacturing (42.64%), wholesale and retail trade (16.64%), and transportation and storage (9.29%) sectors have the largest share of confined space incidents (Fig. 3(c)). This kind of information was obtained by making

uniform the different codes used by each data source for the business activity of companies: the Standard Industrial Classification (SIC) and/or North American Industry Classification System (NAICS) codes used by NIOSH and OSHA, and the Italian classification of economic activities (i. e. ATECO) derived from the European Nomenclature statistique des Activités économiques dans la Communauté Européenne (NACE) considered by INAIL. At this end, we adopted the first level (i.e. the section level) of the International Standard Industrial Classification of all Economic Activities (ISIC) published by United Nations, Department of Economic and Social Affairs (2008).

In addition to the economic activity of the company, we investigated the presence of contractors during the occurrence of the undesired event (Fig. 3(d)). Contractor management is a critical aspect in the safety of confined spaces: the activities performed in such spaces are frequently subcontracted (Burlet-Vienney et al., 2015b), and contractors are usually involved in the majority of fatal incidents (Naghavi Konjin et al., 2019). In the database, 16.20% of incidents involved contractors: about 37.2% of them occurred in business activities related to the construction

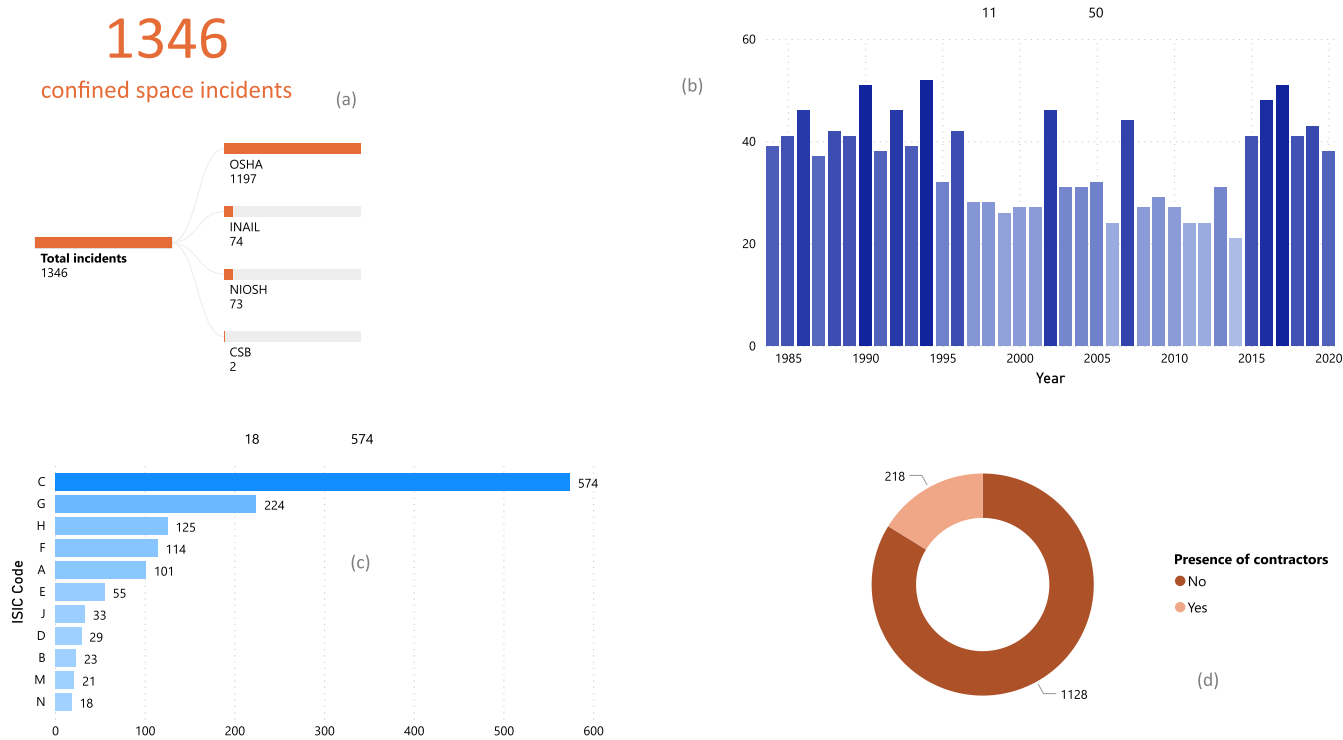


Fig. 3. Incident contextualisation and industry characteristics: (a) total number of incidents in the database and incidents collected from each data source; (b) temporal distribution of the incidents; (c) economic activity of the company; (d) presence of contractors during the incident.

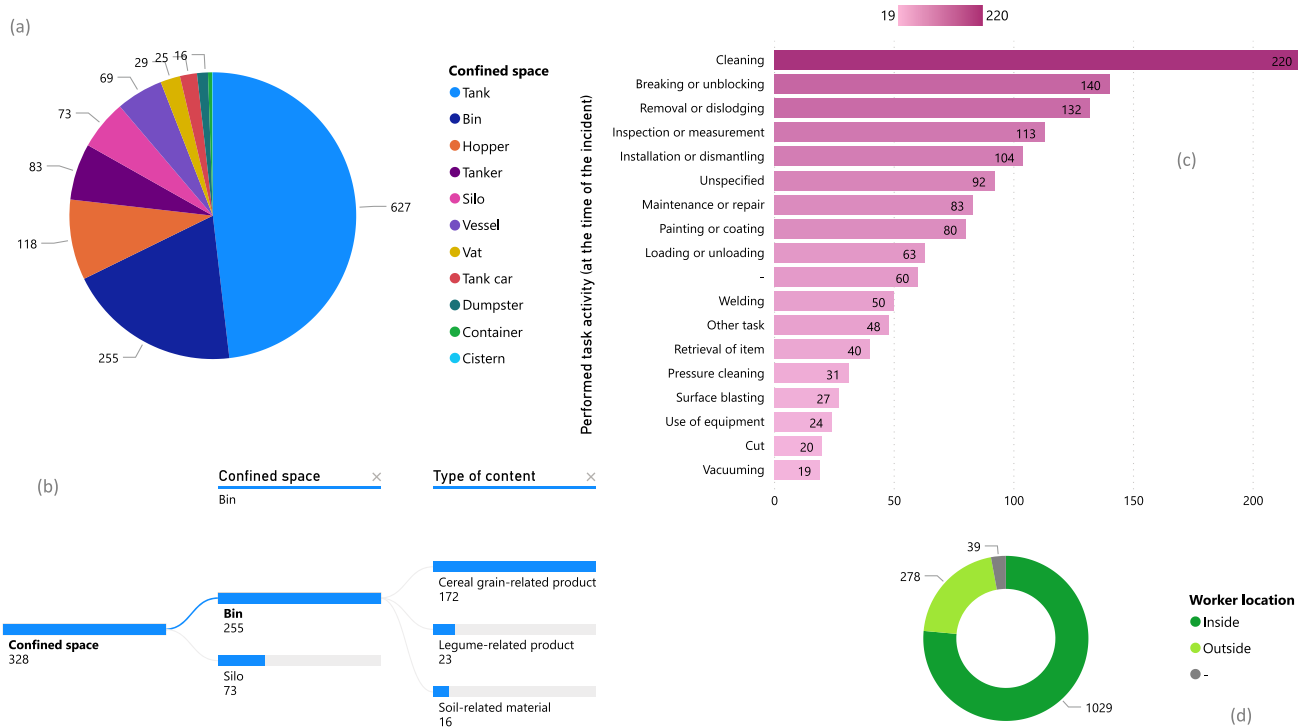


Fig. 4. Confined spaces and worker tasks: (a) incident distribution by the main confined space; (b) type of content in confined spaces; (c) task activities performed at the time of the incident; (d) worker location at the time of the incident.

sector, and about 60% led to fatal consequences for the workers.

3.1.2. Confined spaces and worker tasks

Other features were introduced to characterise the working environments: the confined space and the type of content (besides the

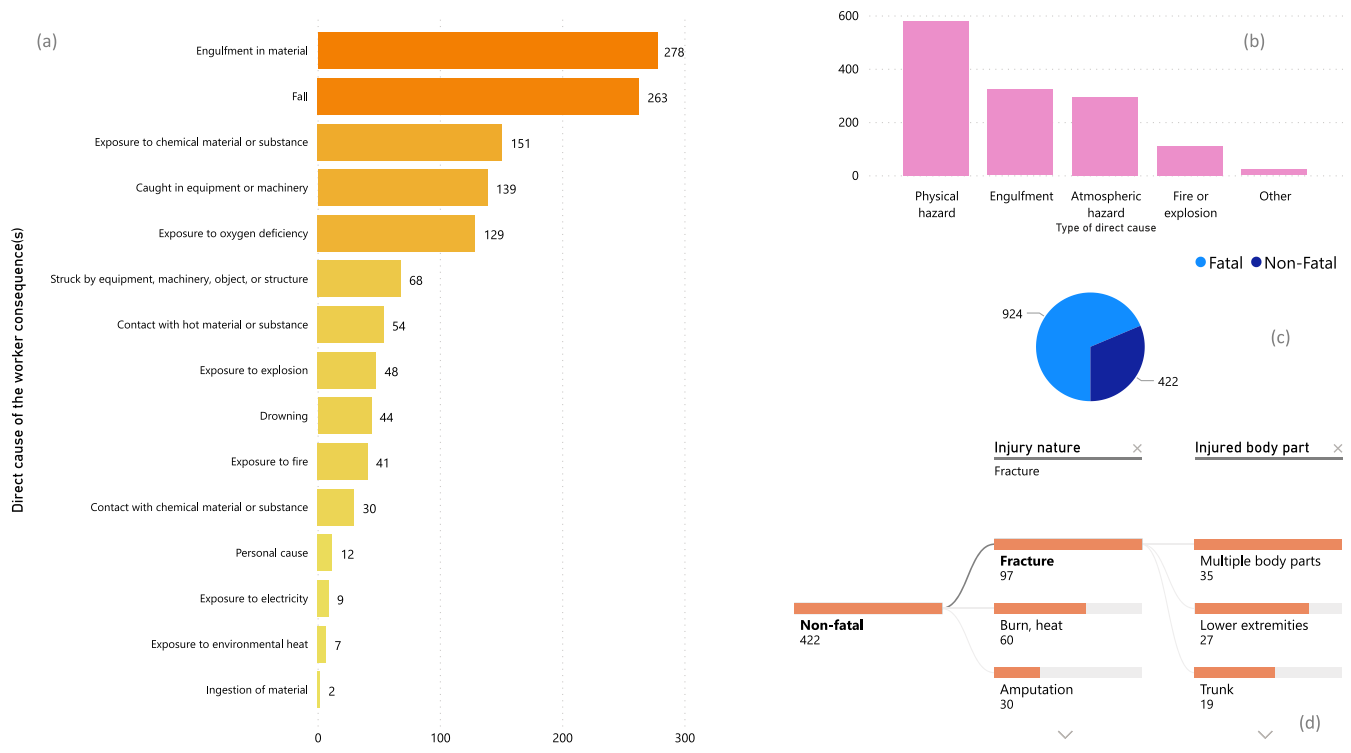


Fig. 5. Incident scenarios and consequences on workers: (a) incident distribution by the main direct cause; (b) incident distribution by the group of direct cause; (c) number of fatal and non-fatal incidents; (d) main injury nature and injured body parts for non-fatal incidents.

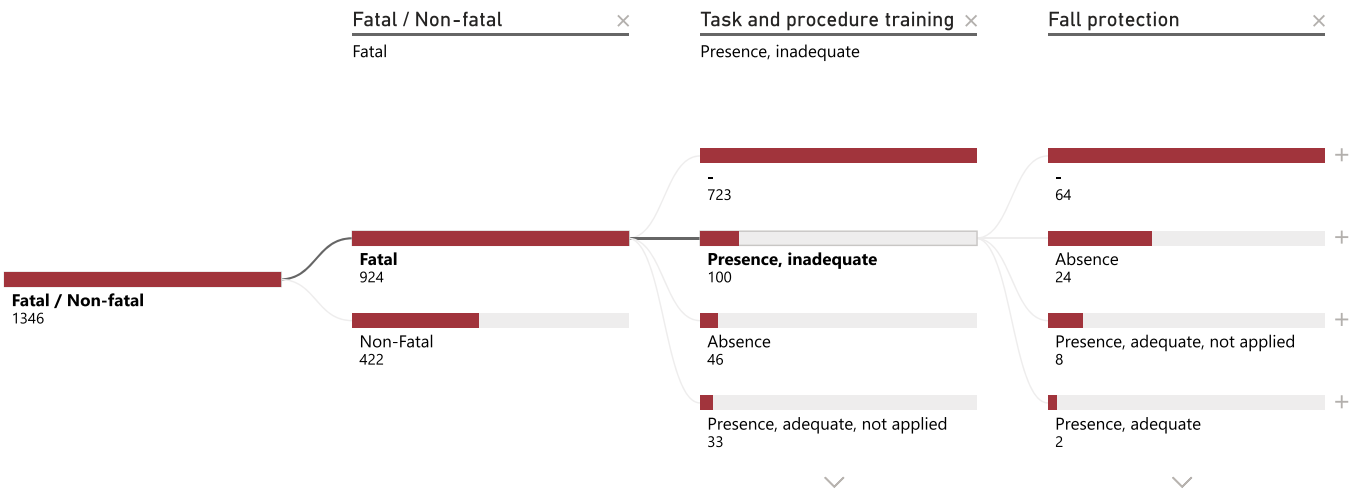


Fig. 6. Safety measures and barriers.

function, i.e. the purpose for which such space is utilised) permit distinguishing the peculiarities among the different types of areas, and giving an indication about their potential hazards (e.g. a mixing tank has a different aim, contains different material in different conditions, and is equipped with different devices or pieces of equipment in comparison to a storage tank or a plating one). Fig. 4(a) emphasises that the collected incidents mainly regarded tank (46.58%), bin (18.95%), and hopper (8.77%). Inside the spaces, different materials or substances (neglecting any equipment or system) can be stored or transported: the content represents an interesting feature for assessing confined space risks because it can be a substance having the potential to produce a hazardous atmosphere or can be a bulk material having the potential for engulfing an entrant (OSHA, 2011). For instance, if pieces of information were available, the bins involved in the incidents mostly contained

cereal grain, legume, or soil -related products (Fig. 4(b)). These contents were frequently responsible for the engulfment in material (164 out of 255 incidents - 64.31%). In such spaces, the most frequent task activities conducted by the workers at the time of the incidents were breaking or unblocking (73 out of 255–28.63%), removal or dislodging (42 out of 255–16.47%), and cleaning (38 out of 255–14.90%). Such activities also represent the traditional ones carried out in all the investigated confined spaces: 492 out of 1346 (36.55%) incidents. Note that cleaning, and removal or dislodging also represent tasks frequently performed by contractors: 26 (11.93%) out of 218 incidents involving contractors regard cleaning operations, 20 (9.17%) concern removal or dislodging. The investigation of the activities conducted by workers at the time of the incident (Fig. 4(c)) plays a relevant role in our study: in addition to the characteristics of the confined space, the

performed task has a direct impact on the overall risk level (Botti et al., 2018, 2022).

The task activities may be performed in or around a confined space (Fig. 4(d)). In the last case, incidents were included in the database if the confined space was a significant factor for the workers' consequences. Most of the selected incidents (about 76.45%) occurred when the employees were inside the confined spaces, while in 20.65% of events the workers were outside. For only a minority of the records (accounting for 2.90% of the total) there are no data or the incident dynamics was unknown (e.g. incident reported in the OSHA data source with ID: 170065767).

3.1.3. Incident scenarios and consequences on workers

Another key feature in the examination of confined space incidents regards the direct cause of the worker consequences, i.e. the loss of human protection, which is the event representing the transfer and interaction of the released energy and/or material to the human target (Stefana et al., 2022b). This feature was detailed in labels referring to both specific events and exposures in OSH and operational safety (e.g. US Department of Labor - Bureau of Labor Statistics, 2012), and groups of causes specifically proposed for the confined space domain (e.g. Selman et al., 2018). Fig. 5(a) shows the single causes most responsible for the worker consequences, while the combined causes (introduced for capturing the potential of interaction among hazards and the creation of synergistic risks as a combination of hazards in confined spaces (Selman et al., 2018)) are poorly represented in the database: only 67 (4.98%) out of 1346 events are characterised by more than one direct cause. Fig. 5(b) displays the frequency of the groups of direct causes. Physical hazards contributed to 44.25% of incidents, engulfment to 25.18% of incidents, atmospheric hazards to 21.89% of incidents, fire or explosion to 6.96%, and other factors to 1.72%.

Engulfment and atmospheric hazards mainly provoked fatal incidents, which also represent the most frequent outcome of the entire database: 924 (68.65%) out of 1346 events were fatal for workers and/or rescuers (Fig. 5(c)). Indeed, confined space incidents often lead to rescue casualties (Chiu et al., 2020), and rescue operations in such spaces frequently cause multiple victims among rescuers during “the chain of solidarity phenomenon” (Botti et al., 2018, 2022). This aspect is stressed also by Selman et al. (2019), who recognised the need for a rescue planning prior to entry and a procedure to reduce the risks in the conduct of a rescue and improve the success rate for the rescue of confined space entrants in difficulty. The results obtained by the analysis of our database confirm these issues: in two incidents, the number of rescuers fatally injured was 4 (the maximum number), which occurred in 1988 in Indiana, US, for a hydrogen cyanide exposure during the cleaning of a storage tank (incident reported in the NIOSH data source with ID: 1988–33), and in 2008 in Italy for a hydrogen sulfide exposure during the cleaning of a tanker (incident reported in the INAIL data source with ID: 1908). The latter represents a milestone incident in confined spaces in Italy, which took place in Molfetta. In both these incidents, the four rescuers died in addition to one worker fatally injured. In a non-fatal event taking place in 1990 in Washington, 18 rescuers were injured (in addition to two workers non-fatally injured) due to a carbon monoxide exposure in a tank: this represents the confined space incident with the maximum number of rescuers non-fatally injured in our database.

All these examples emphasise that incidents with fatal or non-fatal consequences for rescuers mainly regard events with atmospheric causes, and this is also confirmed by considering the entire database. Indeed, 87.85% of incidents involving rescuers were the result of exposure to oxygen deficiency and/or exposure to chemical material or substance. This fact is in agreement with the literature about confined spaces: for instance, Suruda et al. (1994) emphasise that “persons attempting rescue who die in confined spaces most often succumb to atmospheric hazards such as oxygen deficiency or a toxic gas exposure”, and, according to Selman et al. (2019), “rescuer deaths have been found

to be almost exclusively the result of atmospheric hazards”.

For non-fatal incidents, the severity of the consequences is described by means of features about the injury nature and injured body parts, and whose labels were defined on the basis of available and well-known classifications (i.e. the nature of injuries and illnesses proposed by the US Department of Labor - Bureau of Labor Statistics, 2012, and the human body segments provided by the International Standard ISO 11226 (ISO, 2000)). Fig. 5(d) shows a tree visualising the main injury nature and injured body parts: for non-fatal incidents, fracture represented the most frequent type of injuries, which mostly affected multiple body parts (i.e. combination of different injured body parts).

3.1.4. Safety measures and barriers

A set of defined features regard the analysis about the presence, performance, and effectiveness of safety measures and barriers. The term “safety measures” is mainly employed in OSH for identifying the set of actions and controls able to enhance OSH, eliminate hazards, and reduce risks (ISO, 2018b), while the locution “safety barrier” is mostly adopted in major accident and operational safety domain for referring to a physical and/or non-physical means to prevent, control, or mitigate undesired events (Sklet, 2006). We defined a feature for each safety measure or barrier that could have an impact on the confined space incidents, and classified them according to traditional available classifications (e.g. hierarchy of controls (ISO, 2018b), safety functions (Delvosalle et al., 2004)). In addition to the presence or absence of safety measures and barriers during the event scenario, we introduced some details about their performance and effectiveness through proper labels, such as: (1) absence, (2) presence, adequate, (3) presence, adequate, modified, (4) presence, adequate, not applied, and (5) presence, inadequate. The list, description, and classification of the safety measures and barriers defined for this study are reported in Table A2.

Fig. 6 reports an example of the impacts of the task and procedure training, and fall protection systems on the incident outcomes in terms of fatalities or non-fatalities. The most critical issue about these features regards the scarce availability of information and data: the majority of them have a completeness rate lower than 10%, and only 7 out of 28 are characterised by a completeness rate between 10% and 20%. They are: covers and barriers to avoid entry or fall (14%), task and procedure training (19%), confined space entry and procedures (10%), lock out - tag out procedures (19%), attendant (13%), fall protection (18%), and respiratory protection (14%).

3.2. ML analysis

The ML analysis was carried out to predict: (1) the fatal or non-fatal consequences of incidents, and (2) the direct causes of these consequences. For such purpose, we employed supervised classification models on the confined space incident database:

- supervised because both the features (i.e. the independent variables) and the outcomes (i.e. the dependent variables, in our case: fatal or non-fatal consequences for the first learning task, and direct causes of the consequences for the second learning task) of the undesired events are available in the dataset;
- classification since the objective is to predict the categories (or class labels) of the outcome.

The dataset was split into two sets: training data composed of 943 (70%) records, and test data of 403 (30%). The training data were selected by means of Data Sampler in Orange: we implemented it several times and, for each time, we verified that the labels for key features (e.g. confined space, performed task activity, worker location in relation to the confined space, direct cause of the worker consequence, injury nature) are well represented in training and test data to assure a reasonable balance between these data. Furthermore, for every time that we sampled data, we ran LR, RF, and kNN algorithms, and evaluated the

effects on the ML performance metrics.

3.2.1. Prediction model for fatal and non-fatal consequences

The prediction of fatal and non-fatal consequences of an incident is a supervised learning task that can be modelled as a binary classification problem, i.e. a classification with two possible outcomes (Sen et al., 2020). The dependent dichotomous variable is the type of consequence (also known as injury degree or severity) that assumes the values: “fatal” or “non-fatal”. In our database, an incident is fatal if at least one worker or rescuer perished, otherwise the incident is non-fatal. The independent variables are the other features available in our database (Table A1 in Appendix).

Table 3 shows the performance metrics obtained by the three ML algorithms: the RF model outperforms the other two considering all the metrics (highlighted in bold), whereas kNN is the ML model characterised by the lowest results in terms of all performance metrics. Fig. 7 is the confusion matrix obtained by using the RF algorithm: TP and TN refer to correct predictions by the RF model, while FN and FP to wrong predictions. FN (i.e. an actual fatal incident predicted as a non-fatal one) should receive more attention: misclassifying a fatal incident as a non-fatal incident may lead to underestimate the risks and thus to develop insufficient and ineffective safety measures and barriers. Therefore, FN should be minimised in order to avoid a false sense of low severity of some incidental scenarios. For such purpose, a ML model for the prediction of fatal and non-fatal consequences of an incident should be able to maximise the recall metric: it represents the ratio of positive samples correctly detected by the classifier, and is used as a performance indicator when all positive samples need to be identified (Kang and Ryu, 2019).

The model based on RF produces a limited number of FNs. About half of them are records having falls as the direct cause, and the involved workers mainly inside a tank standing on a ladder. For the majority of these records, there are no information and data about the different considered safety measures and barriers. The other fatal incidents misclassified by the RF algorithm as non-fatal regard the following causes: caught in equipment or machinery (3 records), exposure to oxygen deficiency (2 records), struck by equipment, machinery, object, or structure (2 records), exposure to chemical material or substance (1 record), and personal cause (1 record).

A deeper analysis of the RF algorithm also permits identifying the features mostly influencing the target variable (fatal or non-fatal consequence). Fig. 8 displays the 12 features most contributing to the consequence, and the mean and standard deviation of score for each feature, by considering the recall indicator and setting the number of permutations to 5. The most relevant feature for predicting the fatal or non-fatal outcome of an undesired event in a confined space (derived from RF) is the direct cause, followed by the type of content, the performed task activity, and the confined space function. In Fig. 8, there are also some features related to safety measures and barrier: attendant, use of less hazardous machinery or equipment, fall protection, task and procedure training.

3.2.2. Prediction model for direct causes of workers' consequences

The second supervised learning task investigated in this study has the aim to predict (single or group of) direct causes of workers' consequence (i.e. dependent variable), by using as input (independent variables) the features listed in Table A1 (in Appendix), with the exception of number

Table 3

Performance metrics (average over classes) for the prediction of fatal and non-fatal consequences.

Model	AUC	CA	F1	Precision	Recall
LR	0.836	0.792	0.786	0.784	0.792
RF	0.836	0.836	0.834	0.833	0.836
kNN	0.774	0.759	0.755	0.752	0.759

		PREDICTED		Σ
		Fatal	Non-fatal	
ACTUAL	Fatal	TP = 267	FN = 17	284
	Non-fatal	FP = 49	TN = 70	119
Σ		316	87	403

Fig. 7. Confusion matrix obtained with the RF algorithm (number of instances).

of workers and rescuers fatally and non-fatally injured, injury nature (for non-fatal injured workers and rescuers), injured body part (for non-fatal injured workers and rescuers).

Table 4 shows the performance metrics obtained by the three ML algorithms: the LR model outperforms the other two considering all the metrics (highlighted in bold), while kNN is the ML model characterised by the lowest results in terms of all performance metrics. When the target variable is the group of direct cause, all the metrics improve appreciably for all the models.

Focusing on the LR model, the nature of wrong predictions can be studied more in detail. Firstly, we note that all the combined labels related to the direct causes present in the test data are wrongly predicted. The removal of the records characterised by such combined labels from the entire database (67 out of 1346 records) allows slightly increasing the performance metrics of the LR model: AUC achieves a value equal to 0.921, CA to 0.629, precision to 0.611, and recall to 0.629.

Moreover, the LR model predicting the direct cause misclassifies all the causes in the test data poorly represented in the database (e.g. exposure to electricity, exposure to environmental heat, personal cause). Other kinds of misclassifications in the test data can be recognised as follows:

- “exposure to oxygen deficiency” is frequently predicted as “exposure to chemical material or substance”;
- “exposure to fire” is frequently predicted as “exposure to chemical material or substance”;
- “drowning” is frequently predicted as “exposure to oxygen deficiency” or “fall”.

4. Discussion

The reports on safety-related undesired events are valuable resources for learning from the past and unveiling critical issues in the safety and risk management processes. To maximise the lessons learned from such reports, this paper proposes a systematic approach to develop a structured database suitable for effective ML-based analyses: such approach is primarily applied to the confined space topic, and then inductively generalised to be adapted to various kinds of undesired events in OSH and operational safety domains.

4.1. Learning from confined space incidents

The application of our approach to confined space incidents permits highlighting its key steps and potentialities, as well as its main

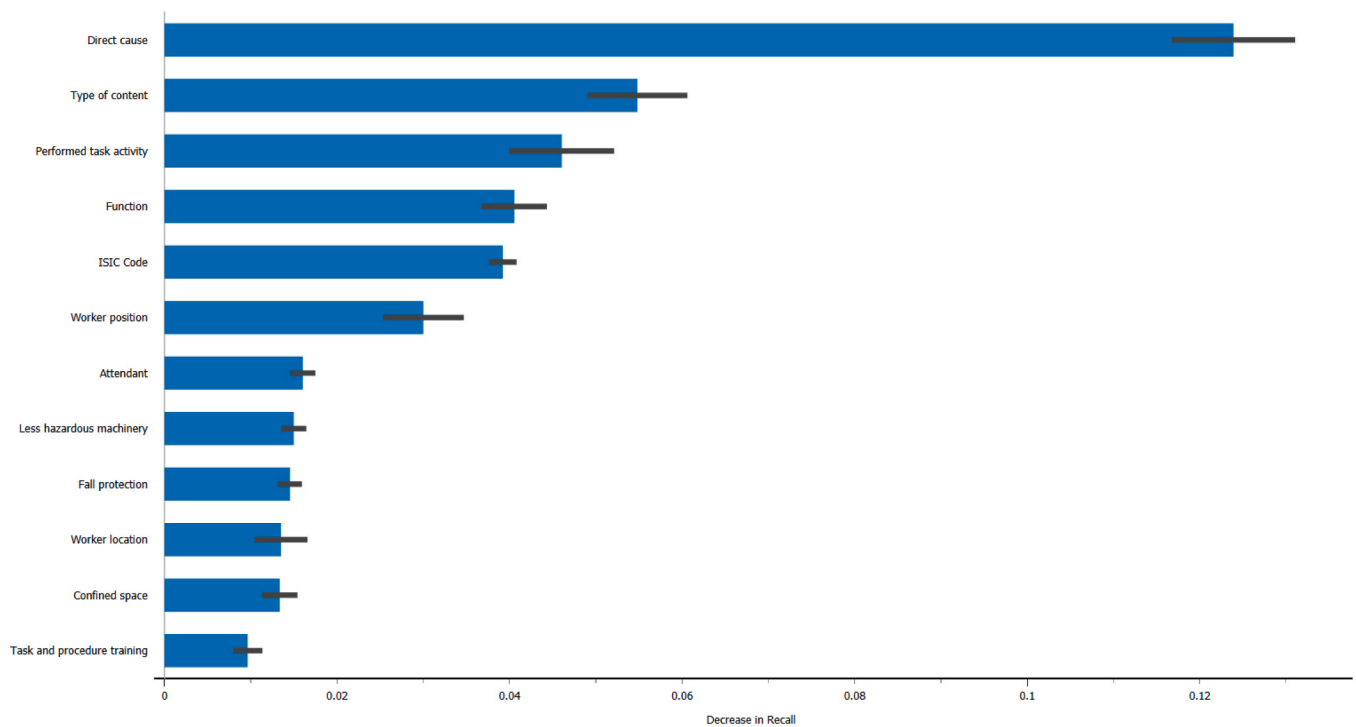


Fig. 8. Feature importance for the RF algorithm.

Table 4

Performance metrics (average over classes) for the predictions of direct causes or their groups.

Target variable	Model	AUC	CA	Precision	Recall
Direct cause	LR	0.922	0.618	0.583	0.618
	RF	0.869	0.571	0.507	0.571
	kNN	0.763	0.429	0.380	0.429
Group of direct cause	LR	0.913	0.774	0.747	0.774
	RF	0.902	0.742	0.723	0.742
	kNN	0.789	0.620	0.586	0.620

difficulties and limitations.

4.1.1. Key steps and potentialities

The entire systematic approach is composed of different steps, from the definition of the purpose of the analysis to the conduction of ML analyses. Such an approach is not linear, but it requires various feedback and iterations among the steps to refine the intermediate and final outcomes. In particular, the record collection and selection step permits better understanding the availability and quality of data and information contained in each data source, testing the inclusion and exclusion criteria previously defined, and identifying clear boundaries for both the analysis and undesired events. Therefore, an effective conduction of such step provides useful inputs to improve the definition and accuracy of the analysis. Then, an enhanced comprehension of the undesired events under investigation is beneficial for the definition of features and labels.

The definition and selection of features represent a key step in our approach. They depend on the objectives and on the desired level of detail of the study, but also on the availability of data and information about the phenomenon under investigation. Therefore, it is necessary to find out a compromise between the need to well describe and characterise the phenomenon and the identification of irrelevant features that can be removed for enabling algorithms to operate faster and more effectively. We structured the confined space database by means of 48 features (each one characterised by specific related labels): they are the

relevant dimensions for the analysis of incident scenarios from the OSH and operational safety perspectives, which permit translating information and data contained in unstructured or semi-structured records from different data sources in a similar and standardised way. The defined features encompass causes, consequences on workers, and safety measures and barriers. To avoid the lack of fundamental pieces of information on crucial features (e.g. direct cause of the worker consequences), we introduced both single and combined labels (refer to Table A1): the combined ones allow specifying all the elements required to completely describe an incident with regard to a specific feature.

The database so created is the main input of the ML analysis. In this study, we focused on a couple of possible supervised classification learning tasks, i.e. the fatal or non-fatal consequences of incidents and the direct cause of these consequences. The prediction of fatal or non-fatal consequences of an incident represents a valuable support for identifying new insights about the most relevant aspects contributing to the injury severity, the most critical scenarios to which to pay attention, the most effective prevention and mitigation measures, and possible additional safety requirements to be defined (Stefana et al., 2022a). In particular, the probability of an incident being fatal or non-fatal largely depends on the direct cause(s), but also on other contributors related to the characteristics of confined spaces (including function and content), the performed activities, and the workers' position and location. This provides a further confirmation about the importance of the characteristics of confined spaces and the conducted activities: safety analysts should take them into account for a comprehensive risk assessment in confined spaces in order to possess all the necessary information and capture their complexity. The possibility to learn the direct causes of the incidents allows comprehending any underlying correlations between a specific type of cause and other incident peculiarities: this could represent a fundamental input for enhancing risk assessment and the entire risk management process in confined spaces (Stefana et al., 2022a).

4.1.2. Main difficulties and limitations

During the application of the approach, we experienced tremendous difficulties during both the selection of pertinent incidents and their

examination to extract relevant data and information. This is linked to the main limitations of the current data sources, also highlighted by [Sarkar et al. \(2020\)](#), [Selman et al. \(2019\)](#), and [Wang and Zhao \(2022\)](#):

- existence of ambiguous descriptions;
- lack of standardised incident data sources;
- existence of different types of data and of unstructured texts demanding an exhaustive human effort in pre-processing for retrieving useful information;
- variability of existing data;
- scarcity of the explanations about the terminology and nomenclature employed in each data source;
- time and effort-consuming process required for the analysis of the reports;
- lack of specific coding for confined space incidents.

The availability of existing data and the variability of the levels of detail among the records indexed in the considered data sources mainly affected the possibility to introduce some other interesting features in the database: e.g. existing regulations and standards for states, workers' citizenship and age, work experience, frequency of entry, temporal evolution of the scenarios, characterisation of delayed consequences on the workers and long-term health effects happening sometime after the incident. For instance, an interesting comparison could be made between records in NIOSH and OSHA. The impossibility of introducing other features could limit the potentialities related to the creation of both an exhaustive database of incidents and ML-based models able to predict further relationships among scenario contributors. This calls for future efforts (in research, but also at political and regulatory levels) about the realisation of standardised data sources collecting and describing incident records precisely, unambiguously, and completely from the OSH and operational safety points of view.

In certain cases, it was quite difficult retrieving precise information about specific relevant features, and some strategies were required to overcome this drawback. For example, to understand the involvement of contractors in incident scenarios (when not precisely reported), we searched for precise words in the descriptions ("contractor" and its synonyms), we identified a set of SIC, NAICS, and ATECO codes specifically related to contractor activities (e.g. SIC: 1799 "Special Trade Contractors, Not Elsewhere Classified"), and/or we compared the business activity codes with the details about the company provided in the records. Another example regards the performed task activities at the time of the incident: if the information was not clearly stated, we referred to the worker occupation.

For some features, we needed to iterate several times the definition of labels: they should be relevant from OSH and operational safety perspectives, but they should not be too numerous to compromise the ML algorithm performance. Therefore, preliminary outcomes from the ML analysis could provide interesting stimuli to refine the identified labels. An example of this in the confined space incident database is related to the type of content. Indeed, we defined its labels by grouping the possible substances and/or materials in homogeneous classes (e.g. cereal grain-related product, food or beverage, metal material), and by classifying chemicals based on reactive groups, i.e. "categories of chemicals that typically react in similar ways because they are similar in their chemical structure" (CAMEO Chemicals). However, this level of detail produced more than 100 labels that negatively affected the performance metrics of prediction models: consequently, we considered only the most interesting chemical clusters from the OSH and operational safety perspectives and/or the most frequent ones as labels about chemical content. Additional classifications about chemical materials or substances (e.g. hazard statements defined by [European Parliament and Council \(2008\)](#) and recently updated by [European Commission \(2022\)](#)) could be considered for future studies about confined spaces.

The above difficulties caused about one year of work for developing the database, during which we constantly discussed divergences about

the record selection and feature extraction, with the aim of reducing uncertainties in the process, increasing the data quality of the entire analysis, and improving the overall database quality. The management of all these aspects is fundamental for minimising the negative repercussions linked to the well-known "garbage in – garbage out" effect, and maximising the possible and reliable outcomes that may be derived from the process of learning from incidents.

In addition to the above difficulties, there are certain limitations on the created database, which are strictly dependent on the selection of data sources. For instance, the temporal and geographical distributions of the incidents should be interpreted with caution because they cannot be considered indicative of the actual worldwide trends of confined space incidents. This was also related to other aspects: (1) the incident collection and coverage of the considered data sources are not homogeneous, (2) the development and, in some cases, the revision of some regulations and standards about confined spaces could have stimulated a different attention and sensitivity towards this topic, and (3) the level of details and the data availability in the records could be slightly different over the years, influencing their inclusion or exclusion in the database. In terms of geographical distribution, there is also an additional issue: the majority of the reports derive from the OSHA, NIOSH, and CSB data sources, and thus most incidents occurred in US states (including federal districts and other territories), and only 74 out of 1346 took place in Italy (the ones recorded in the INAIL data source reporting incidents from 2002 to 2020).

Other limitations concern the results obtained by means of ML analyses. On one side, the rate of data availability and the different values of completeness among the features could influence the feature importance in the prediction models. In the confined space incident database, this is particularly noticeable in the case of safety measures and barriers, which are characterised by different completeness levels that never exceed 20%. On the other side, the performance of the models predicting the direct cause improves when the testing of the ML algorithms were carried out on training data: this could be a signal of overfitting, which occurs when a model fits the training data precisely and fails to generalise on new data ([ISO and IEC, 2022](#)). Such phenomenon could be mainly caused by the following issues: 1) the training data do not contain sufficient data samples and examples to accurately learn all possible labels, and 2) there are too many features that make the model complex. The overfitting problem could be inhibited by expansion of the training data and/or regularisation strategies ([Ying, 2019](#)). The former can be realised by acquiring new data to provide sufficient samples for learning ([Ying, 2019](#)): in our study, for instance, we could collect records contained in other European and international incident data sources (e.g. French Institut National de Recherche et de Sécurité (INRS) archives) that were not considered here for the use of languages different from English and Italian, or we could include other kinds of confined spaces (e.g. crawl spaces in machinery, underground tunnels, pits). The latter can be done by selecting only the useful features or minimising the weights of the features that have little influence on the final classification in order to limit the number of features taken into consideration by the model ([Ying, 2019](#)). In our research, regularisation strategies could be easily implemented thanks to both the limited computation times offered by Orange capabilities and the versatility of our database and approach. Indeed, we designed and developed our database in such a way that some features can be neglected, while others further detailed in order to build increasingly accurate prediction models. For instance, an increase of the size of the training data, without extending the current database and maintaining the same set of features, could improve the performance of the LR model if the training data are set to 80% of the entire database: AUC achieves a value equal to 0.926, CA to 0.625, precision to 0.614, and recall 0.625.

4.2. Learning from safety-related undesired events

The application of the systematic approach to the confined space

incidents allows learning from the process itself, enhancing the overall methodology, and inductively generalising it for being adopted in the analysis of undesired events in OSH and operational safety domains.

4.2.1. Generalised systematic approach for database development and ML analyses

Fig. 9 proposes the generalised systematic approach for database development and ML analyses, whose steps are detailed in the following paragraphs. Note that each step is necessary for performing the following ones, but, at the same time, it could be improved and enhanced thanks to the knowledge acquired during the process: throughout the approach, there are continuous feedback and iterations that refine the intermediate and final outcomes of the analysis. Therefore, the approach provides a reliable support for critical decision-making in OSH and operational safety, whose proper adoption requires the human involvement of safety analysts (Paltrinieri, 2022).

4.2.1.1. Define the purpose of the analysis and undesired events. The purpose of the analysis and undesired events should be clearly defined at the beginning of the study, with regard to both OSH and operational safety perspectives, and the users' requirements. Indeed, fitness for use, and requirements and needs of the users should be clearly identified since they drive the entire database design, development, and use process (Goasdoué et al., 2007; Hoxmeier, 1998).

Reviews of scientific literature, technical documents, and previous statistical analyses could help refine and restrict the interest towards a precise set of safety-related undesired events, and increase the knowledge about their hazards, risks, critical scenarios, and typical working environments. During this step, the main keywords used for describing the undesired events should be identified, and a set of inclusion and exclusion criteria established to precisely delineate the boundaries of the research. Such criteria should be specified at the beginning of the analysis, and continuously enhanced during the entire approach to assure a certain level of accuracy and consistency of the database. Inclusion and exclusion criteria should also address the data quality issue: the records containing ambiguous descriptions about the mechanism and scenario of an undesired event should be excluded from the analysis.

4.2.1.2. Characterise available data sources. The definition of the purpose of the analysis and the undesired events allows searching for

relevant available data sources. These data sources can be freely accessible online, available upon request, and/or commercially available, in different formats (e.g. Web pages, Microsoft® Excel spreadsheets, PDF files), and in various languages. It is advisable to consider both national and international data sources, the most updated ones, and those most frequently cited in the scientific literature and technical documents. Interesting reports should be gathered through data sources that are specifically related to the investigated phenomenon, as well as generic sources collecting different kinds of safety-related undesired events.

An analysis about the main characteristics (e.g. the process related to data collection, the involved actors, the purposes) and the quality of the identified data sources should be conducted in order to select only those that guarantee good or high-quality event reports. Finally, the characterisation of the selected data sources could also better frame the undesired events of interest, refine the used keywords, and stimulate a review of the inclusion and exclusion criteria.

4.2.1.3. Collect and select records. The records contained in each data source should be collected in a database with a user-friendly format so that all available pieces of information are easily visualised and can be analysed, and filtering and cleaning operations can be carried out. In particular, the records should be analysed in accordance with the inclusion and exclusion criteria to select only those relevant for the purpose of the analysis: the researchers should independently examine a subset of records and discuss any doubts and/or uncertainties about their inclusion / exclusion during proper brainstorming sessions. This allows increasing the confidence level about the reliability of the results.

At the end of the step related to the record analysis and selection, cleaning operations on each database should be performed. Specifically, a deduplication step assures the removal of any possible duplicates.

4.2.1.4. Update with new reports. The database should not be static, but it should be dynamically updated with new undesired event reports when available. Therefore, a strategy for constantly monitoring the presence of new records in the data sources and for automatically adding them to the database should be designed.

4.2.1.5. Define features and labels. After the record collection and selection step, features and corresponding labels should be defined. An

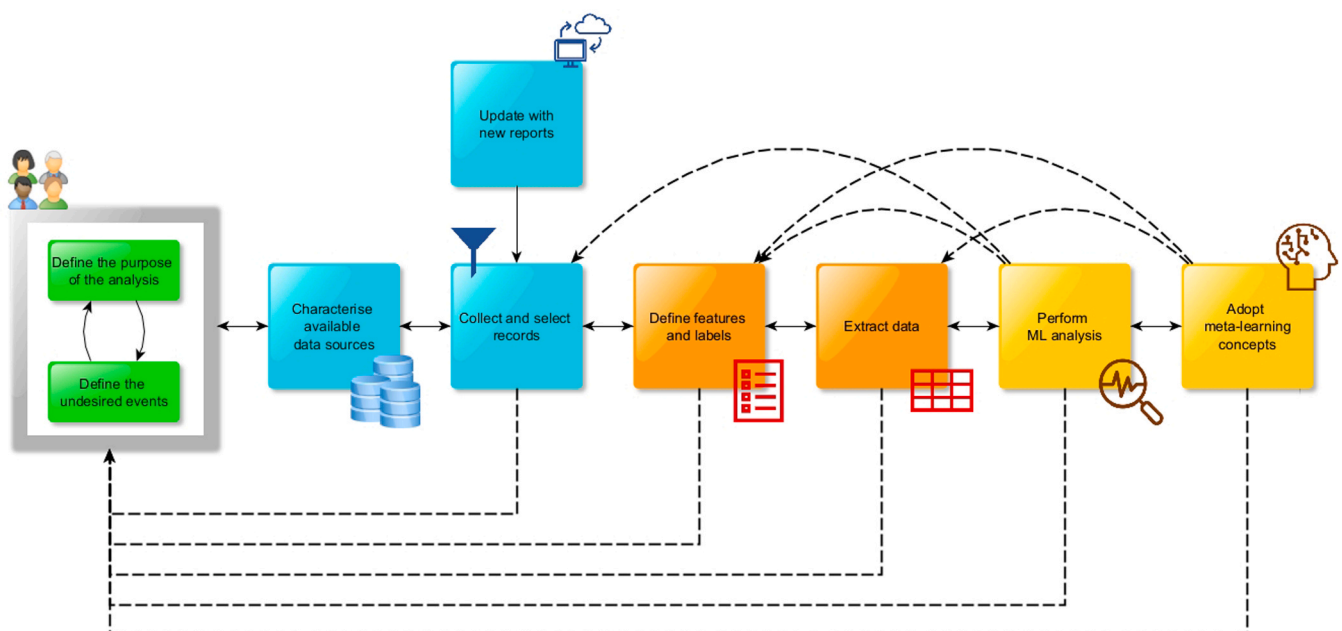


Fig. 9. Generalised systematic approach for database development and ML analyses.

overview of potential valuable features for undesired events in OSH and operational safety is proposed in [Table 5](#).

Potential features of interest could be derived from the use of mathematical models to estimate relevant parameters characterising the direct cause of the workers' consequences (e.g. [Stefana et al., 2021b](#); [Yan et al., 2023](#)) and/or consequence assessment approaches to identify and quantify adverse consequences and impacts from an hazardous event (e.g. [Arunraj and Maiti, 2009](#)). This could represent an interesting step towards the integration between physics-based and data-driven approaches in the OSH and operational safety fields. Other features could regard other events of the scenario to capture consecutive risks or domino effects.

Furthermore, a set of features related to safety measures and barriers should be included in order to understand their role in the occurrence, evolution, and/or mitigation of undesired events. To capture all the means able to treat risks in OSH and operational safety, we prefer referring to both “safety measures” and “safety barriers”. The labels related to the safety measures and barriers should be able to characterise their presence, performance, and effectiveness. The explanation of the actual safety measures and barriers adopted at the time of the event, along with their implementation status could provide relevant inputs for the process of learning from undesired events. For this reason, the records with the details about the safety measures and barriers could

Table 5
Overview of potential features for structuring a safety database.

Group of features	Feature
Identifier	ID Original data source
Event contextualisation	Year State
Industry	Business activity of the company where the event happened
Working environment of the event	Department Specific area Dimensions of the environment Internal conditions (temperature, pressure, air composition) Substances present Equipment present
Workers	Occupation Qualification Presence of contractors Age Nationality Gender
Task analysis	Activities performed by workers during the event Workers' location Operational status of the plant / equipment
Causes of the event	Direct cause of the workers' consequence Secondary causes Contributory factors Release flow rate Substance concentration ...
Consequences of the event	Presence of domino effects Number of workers fatally injured Number of workers non-fatally injured Injury nature Injured body part Number of persons (other than workers) fatally injured Number of persons (other than workers) non-fatally injured Damage to equipment Damage to environment Economic damage
Safety measures and barriers	Safety measure and barrier A Safety measure and barrier B ... Safety measure and barrier N

receive more emphasis and significance compared to other instances by means of the assignment of different weights.

To effectively define features and labels, we suggest considering all the characteristics relevant for the undesired events under investigation from the OSH and operational safety perspectives, try to extract data about them from the reports, and then select the features with a reasonable level of completeness. In addition, relevant international classifications about specific features (e.g. International Labour Organization (ILO), and/or World Health Organization (WHO) for diseases) may help to structure a standardised database.

4.2.1.6. Extract data. Different strategies could be implemented for data extraction and label assignment. In particular, text mining processes and NLP methods (e.g. tokenization, clustering) could be adopted to extract information and knowledge from unstructured or semi-structured text when a database is composed of thousands and thousands of records. When the structure and peculiarities of the reports (refer for instance to the challenging tasks mentioned by [Khurana et al., 2023](#)) could limit the opportunities offered by NLP and decrease its performance, a manual label assignment could be considered. However, a combination of them appears a good compromise between precision and the required human efforts.

The data extraction step could also help introducing additional features and labels and/or refining the ones previously defined. Finally, the missing values in the database should be replaced with a standardised term or symbol. This last step permits obtaining the final version of the structured database with an indication about the completeness level for each feature. Such database could then be imported in existing software tools dedicated to dynamic data visualisation that support different types of analyses of the features and labels. This could support a real-time and continual monitoring of the trends of features and labels when new reports are provided.

4.2.1.7. Perform ML analysis. The learning process is grounded on the employment of ML principles. In this step, a proper selection of techniques and software for performing ML analyses should be carried out.

The literature offers a wide spectrum of supervised and unsupervised learning techniques that could be implemented for extracting knowledge from safety-related undesired event reports. For instance, supervised regression and classification models permit predicting relevant features describing an event scenario (e.g. causes or consequences). Alternative ML algorithms should be implemented to build prediction models, whose performance metrics could be examined and compared to identify the most suitable one(s) for the investigated task.

Furthermore, when the instances are unlabelled, unsupervised learning could be adopted ([Kotsiantis et al., 2006](#)). Among its possible techniques, clustering seems to be a promising one: it groups a given collection of unlabelled patterns into meaningful clusters ([Jain et al., 1999](#)). In the undesired event analysis, ML clustering techniques may provide a support during the classification and selection of records, as well as the definition of features and labels. The research could also include Multi-Task Learning (MTL), which is a specific learning paradigm in ML aiming “to learn multiple related tasks jointly so that the knowledge contained in a task can be leveraged by other tasks” ([Zhang and Yang, 2022](#)). MTL permits combining different learning tasks to improve the generalisation performance of all the tasks.

4.2.1.8. Adopt meta-learning concepts. The entire approach could be enhanced by exploiting meta-learning concepts: meta-learning refers to the process of acquiring and using knowledge about learning to understand and improve the performance of learning systems, and allows accumulating experience on the performance of multiple applications of a learning system ([Stefana et al., 2023](#)). The meta-learning implementation could enhance a more effective NLP application in order to easily update the database with new records, filter relevant information,

and reduce human labour for labelling. In addition, the adoption of meta-learning methods could help in extracting effective meta-features and recommending the best ML algorithm to predict risks in specific sets of undesired events, also in presence of a limited amount of data and small sample size (Li et al., 2021).

5. Conclusions

This paper enhances the learning from undesired events in OSH and operational safety by developing a systematic approach for creating a structured database suitable for effective ML-based analyses. This approach was primarily applied to confined space incidents that were collected from various and separate national and international data sources in a unique database. Such database was composed of more than 1300 fatal and non-fatal incidents that occurred in US and Italy from 1984 and 2021. Each incident is described through 48 features and related labels, which are defined by considering the peculiarities of confined spaces and the risk management process in OSH and operational safety. Some models based on three well-known ML techniques (i. e. LR, RF, kNN) were built to classify and predict the fatal or non-fatal consequences of incidents and their direct causes. This application allowed overcoming the current limited use of ML techniques in the process of learning from confined space incidents.

Finally, a generalised approach was inductively proposed to learn

from various kinds of undesired events in OSH and operational safety domains. It is composed of several steps, including data source characterisation, record collection and selection, definition of appropriate features and labels, data extraction, and ML analysis. It is not a linear sequence of steps, but it requires various feedback and iterations to continuously refine the intermediate and final outcomes. Such an approach is domain-independent, and provides a novel support for maximising the lessons learned from unstructured reports of incidents and accidents.

Future research activities could be focused on both the enhancement and application of the generalised systematic approach to other safety-related undesired events. For instance, possible strategies to easily update the database should be designed and tested. The combined use of NLP and meta-learning is an interesting topic that could deserve future investigation for managing narrative event descriptions. Finally, the approach could be applied to analyse other types of undesired events for testing its general validity, and extending and/or refining the set of potential features proposed.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

Table A1

Features and labels for the analysis of confined space incidents (when relevant, their details are in brackets), and percentage of completeness in the database.

Feature	Labels	% of completeness
	Single (in alphabetical order) Combined	
Record source	(1) CSB; (2) INAIL; (3) NIOSH; (4) OSHA	No 100%
Year	(1) 1984; (2) 1985; (3) 1986; (4) 1987; (5) 1988; (6) 1989; (7) 1990; (8) 1991; (9) 1992; (10) 1993; (11) 1994; (12) 1995; (13) 1996; (14) 1997; (15) 1998; (16) 1999; (17) 2000; (18) 2001; (19) 2002; (20) 2003; (21) 2004; (22) 2005; (23) 2006; (24) 2007; (25) 2008; (26) 2009; (29) 2010; (30) 2011; (31) 2012; (32) 2013; (33) 2014; (34) 2015; (35) 2016; (36) 2017; (37) 2018; (38) 2019; (39) 2020; (40) 2021	No 100%
State (including federal districts and other territories)	(1) Alabama; (2) Alaska; (3) Arizona; (4) Arkansas; (5) California; (6) Colorado; (7) Connecticut; (8) Delaware; (9) District of Columbia; (10) Florida; (11) Georgia; (12) Hawaii; (13) Idaho; (14) Illinois; (15) Indiana; (16) Iowa; (17) Italy; (18) Kansas; (19) Kentucky; (20) Louisiana; (21) Maine; (22) Maryland; (23) Massachusetts; (24) Michigan; (25) Minnesota; (26) Mississippi; (27) Missouri; (28) Montana; (29) Nebraska; (30) Nevada; (31) New Hampshire; (32) New Jersey; (33) New Mexico; (34) New York; (35) North Carolina; (36) North Dakota; (37) Ohio; (38) Oklahoma; (39) Oregon; (40) Pennsylvania; (41) Puerto Rico; (42) Rhode Island; (43) South Carolina; (44) South Dakota; (45) Tennessee; (46) Texas; (47) Utah; (48) Vermont; (49) Virgin Islands; (50) Virginia; (51) Washington; (52) West Virginia; (53) Wisconsin; (54) Wyoming	No 100%
ISIC code	(1) A (i.e. Agriculture; forestry and fishing); (2) B (i.e. Mining and quarrying); (3) C (i.e. Manufacturing); (4) D (i.e. Electricity; gas, steam and air conditioning supply); (5) E (i.e. Water supply; sewerage, waste management and remediation activities); (6) F (i.e. Construction); (7) G (i.e. Wholesale and retail trade; repair of motor vehicles and motorcycles); (8) H (i.e. Transportation and storage); (9) I (i.e. Accommodation and food service activities); (10) J (i.e. Information and communication); (11) L (i.e. Real estate activities); (12) M (i.e. Professional, scientific and technical activities); (13) N (i.e. Administrative and support service activities); (14) O (i.e. Public administration and defence; compulsory social security); (15) P (i.e. Education); (16) Q (i.e. Human health and social work activities); (17) R (i.e. Arts, entertainment and recreation)	No 100%
Presence of contractors	(1) No; (2) Yes	No 100%
Confined space	(1) Bin; (2) Cistern; (3) Container; (4) Dumpster; (5) Hopper; (6) Hopper car; (7) Silo; (8) Tank; (9) Tank car; (10) Tanker; (11) Vat; (12) Vessel	Yes, combination of single labels 100%
Function	(1) Anodising; (2) Batch; (3) Brew; (4) Cleaning; (5) Coating; (6) Condensate; (7) Converter; (8) Cooling; (9) Crystalliser; (10) Deaerator; (11) Degreasing; (12) Demineraliser; (13) Digester; (14) Dipping; (15) Discharge; (16) Dissolving; (17) Distillation; (18) Drainage; (19) Drying; (20) Dump; (21) Economiser; (22) Extractor; (23) Fermentation; (24) Filling; (25) Filtration; (26) Flocculation; (27) Fractionation; (28) Fracturing; (29) Furnace; (30) Grinder; (31) Heat treating; (32) Incinerator; (33) Knockdown; (34) Loading; (35) Melting; (36) Mixing; (37) Neutralising; (38) Pickling; (39) Plating; (40) Pressure; (41) Processing; (42) Pump; (43) Purifying; (44) Quench; (45) Reactor; (46) Recarbonation; (47) Receiving; (48) Recovery; (49) Recycling; (50) Reduction; (51) Rinse; (52) Scrubber; (53) Septic; (54) Settling; (55) Skimmer; (56) Slugging; (57) Sparkling; (58) Steam; (59) Storage; (60)	Yes, combination of single labels 47%

(continued on next page)

Table A1 (continued)

Feature	Labels		% of completeness
	Single (in alphabetical order)	Combined	
Type of content	Stripping; (61) Sump; (62) Testing; (63) Tote; (64) Transformer; (65) Transition; (66) Transport; (67) Treatment; (68) Turbo-mist spray; (69) Weight (1) Adhesive, insulation, paint, or ink (e.g. latex, residual bee wax caps, water-based latex paint); (2) Animal food (e.g. feed, pet food mixture, dog food, raw scrap dog food material); (3) Catalyst; (4) Cereal grain-related product (e.g. grain, rice, corn, shelled corn, corn screening, milo, wheat); (5) Chemical (i.e. chemical substances with the exception of acids, bases, and hydrocarbons); (6) Chemical, acid; (7) Chemical, base; (8) Chemical, hydrocarbon; (9) Cotton-related product (e.g. cotton seed); (10) Decomposed animal part (e.g. chicken blood, scrap chicken parts, decomposed hog entrails); (11) Detergent or soap; (12) Dye and pigment; (13) Food or beverage (e.g. chocolate, dough, frozen food, wine, beer, whey); (14) Forage or grass-related product (e.g. crop, haylage, silage, grass seed, fescue seed); (15) Glass-related product; (16) Legume-related product (e.g. soya bean, bean, alfalfa, peanut hull); (17) Metal material (e.g. ferrous flake, slag, metal scrap, molten steel); (18) Mixture of soil-related materials (e.g. asphalt, concrete, cement, slurry, cement powder); (19) Molasses; (20) Other content (e.g. alpha amylase); (21) Paper-related product; (22) Plastic-related product; (23) Plating and anodising solution; (24) Rubber-related product; (25) Sewage or sludge (e.g. sludge, wastewater, sewage, septic water, liquid waste, slush, mud); (26) Soil-related material (e.g. compost, fertiliser, sand, stone, gravel, mulch, soil); (27) Unspecified material or substance (e.g. liquid, dust, material, residue, hazardous substance, debris); (28) Water-related product (e.g. water, ice, steam); (29) Wood-related product (e.g. wood chip, sawdust, wood logs, wood log)	Yes, combination of labels indicated as “multiple”	78%
Performed task activity	(1) Breaking or unblocking (e.g. knocking, chipping, unclogging, clearing a blockage); (2) Cleaning; (3) Cut (e.g. cutting, flame-cutting); (4) Inspection or measurement (e.g. checking, inspecting, monitoring, measuring, gauging, determining, sampling); (5) Installation or dismantling (e.g. positioning, installing, taking down, unplugging, erecting a scaffold); (6) Loading or unloading (e.g. pouring, transferring, emptying); (7) Maintenance or repair (e.g. making repairs, changing, adjusting); (8) Other task (e.g. walking, watching someone, waiting); (9) Painting or coating (e.g. spraying, lining the inside, relining the interior); (10) Pressure cleaning (e.g. pressure washing, power washing); (11) Removal or dislodging (e.g. scooping, sweeping, shovelling, scraping); (12) Retrieval of item; (13) Surface blasting (e.g. sandblasting, hydro-blasting, water blasting, abrasive blasting); (14) Unspecified; (15) Use of equipment (e.g. turning on/off, grinding, operating a machine); (16) Vacuuming (e.g. sucking, operating a vacuum hose); (17) Welding (e.g. performing gas shielding)	No	96%
Worker location in relation to the confined space	(1) Inside; (2) Outside	No	97%
Worker position	(1) At the hatch or opening (including gates or holes); (2) In the upper part of the confined space (including areas near the top); (3) On a ladder (e.g. suspended, fixed, extension, portable); (4) On a lift device (including an aerial lift or a boatswain’s chair); (5) On a plank, board, or beam (i.e. wood support, including deck, piece of plywood, pallet, or stool); (6) On a railing or barrier guard; (7) On a scaffold, platform, or catwalk (including tray, plate, internal roof, internal cover, walkway, walking surface, or rope); (8) On a top surface of the confined space (including a gate, manhole, cover, grate, lid, or roof); (9) On the edge of the confined space; (10) On the floor; (11) On the material surface	Yes, combination of single labels	39%
Direct cause of the worker consequences	(1) Bodily motion (including overexertion related to pushing, pulling, holding, or repetitive tasks); (2) Caught in equipment or machinery (including gates, openings, running and operating equipment); (3) Contact with chemical material or substance; (4) Contact with cold material or substance; (5) Contact with hot material or substance; (6) Drowning; (7) Engulfment in material; (8) Exposure to change in water pressure; (9) Exposure to chemical material or substance (including toxic, irritant, corrosive airborne substances); (10) Exposure to electricity; (11) Exposure to environmental cold; (12) Exposure to environmental heat; (13) Exposure to explosion; (14) Exposure to fire; (15) Exposure to oxygen deficiency (including atmosphere with not chemically reactive substances and created for fermentation processes); (16) Fall (including fall on same level and to lower level); (17) Ingestion of material; (18) Personal cause (including heart attack, myocardial infarction, ischemic heart disease, asthma attack); (19) Struck by equipment, machinery, object, or structure (including collapsing structures or materials, falling or flying object or equipment part, discharged object or substance) Alternatively: (1) Atmospheric hazard (including exposure to chemical material or substance, exposure to oxygen deficiency); (2) Fire or explosion (including exposure to explosion, exposure to fire); (3) Engulfment (including engulfment in material, drowning); (4) Physical hazard (including caught in equipment or machinery, contact with cold material or substance, contact with chemical material or substance, contact with hot material or substance, exposure to electricity, fall, ingestion of material, struck by equipment, machinery, object, or structure); (5) Other (including bodily motion, exposure to change in water pressure, exposure to environmental cold, exposure to environmental heat, personal cause)	Yes, combination of single labels	100%
Number of workers fatally injured	Numerical	No	100%
Number of rescuers fatally injured	Numerical	No	100%
Number of workers non-fatally injured	Numerical	No	100%

(continued on next page)

Table A1 (continued)

Feature	Labels	% of completeness
	Single (in alphabetical order) Combined	
Number of rescuers non-fatally injured	Numerical	No 100%
Injury nature (for non-fatal injuries)	(1) Amputation; (2) Asphyxia; (3) Burn, chemical; (4) Burn, cryogenic; (5) Burn, electrical; (6) Burn, friction; (7) Burn, heat; (8) Concussion; (9) Contusion; (10) Cut, laceration; (11) Dislocation; (12) Electrocution, electric shock; (13) Embolism, air or fat; (14) Fracture; (15) Frostbite; (16) Heat exhaustion; (17) Paralysis; (18) Poisoning (i.e. injury caused by an exposure to toxic substance); (19) Sprain, strain, tear; (20) Suffocation; (21) Surface wound and bruise; (22) Unspecified disease and disorder; (23) Unspecified traumatic injury and disorder	Yes, combination of single labels 100%
Injured body part (for non-fatal injuries)	(1) Body system (i.e. various systems of the body and their functioning); (2) Head; (3) Lower extremities; (4) Neck; (5) Trunk; (6) Upper extremities	Yes, combination of labels indicated as “multiple” 96%
Injury nature (for non-fatal injured rescuers)	(1) Amputation; (2) Asphyxia; (3) Burn, chemical; (4) Burn, cryogenic; (5) Burn, electrical; (6) Burn, friction; (7) Burn, heat; (8) Concussion; (9) Contusion; (10) Cut, laceration; (11) Dislocation; (12) Electrocution, electric shock; (13) Embolism, air or fat; (14) Fracture; (15) Frostbite; (16) Heat exhaustion; (17) Paralysis; (18) Poisoning (i.e. injury caused by an exposure to toxic substance); (19) Sprain, strain, tear; (20) Suffocation; (21) Surface wound and bruise; (22) Unspecified disease and disorder; (23) Unspecified traumatic injury and disorder	Yes, combination of single labels 100%
Injured body part (for non-fatal injured rescuers)	(1) Body system (i.e. various systems of the body and their functioning); (2) Head; (3) Lower extremities; (4) Neck; (5) Trunk; (6) Upper extremities	Yes, combination of labels indicated as “multiple” 100%
Devices minimising the need for entry	(1) Absence; (2) Presence, adequate; (3) Presence, adequate, modified; (4) Presence, adequate, not applied; (5) Presence, inadequate	No 2%
Use of less hazardous substances	(1) Absence; (2) Presence, adequate; (3) Presence, adequate, modified; (4) Presence, adequate, not applied; (5) Presence, inadequate	No 5%
Use of less hazardous machinery or equipment	(1) Absence; (2) Presence, adequate; (3) Presence, adequate, modified; (4) Presence, adequate, not applied; (5) Presence, inadequate	No 6%
Ventilation systems	(1) Absence; (2) Presence, adequate; (3) Presence, adequate, modified; (4) Presence, adequate, not applied; (5) Presence, inadequate	No 8%
Purging of the atmosphere	(1) Absence; (2) Presence, adequate; (3) Presence, adequate, modified; (4) Presence, adequate, not applied; (5) Presence, inadequate	No 4%
Equipment or position guarding	(1) Absence; (2) Presence, adequate; (3) Presence, adequate, modified; (4) Presence, adequate, not applied; (5) Presence, inadequate	No 9%
Covers and barriers to avoid entry or fall	(1) Absence; (2) Presence, adequate; (3) Presence, adequate, modified; (4) Presence, adequate, not applied; (5) Presence, inadequate	No 14%
Lighting systems	(1) Absence; (2) Presence, adequate; (3) Presence, adequate, modified; (4) Presence, adequate, not applied; (5) Presence, inadequate	No 3%
Isolation and containment systems	(1) Absence; (2) Presence, adequate; (3) Presence, adequate, modified; (4) Presence, adequate, not applied; (5) Presence, inadequate	No 1%
Interlock systems	(1) Absence; (2) Presence, adequate; (3) Presence, adequate, modified; (4) Presence, adequate, not applied; (5) Presence, inadequate	No 1%
Testing equipment	(1) Absence; (2) Presence, adequate; (3) Presence, adequate, modified; (4) Presence, adequate, not applied; (5) Presence, inadequate	No 2%
Equipment maintenance	(1) Absence; (2) Presence, adequate; (3) Presence, adequate, modified; (4) Presence, adequate, not applied; (5) Presence, inadequate	No 5%
Confined space training	(1) Absence; (2) Presence, adequate; (3) Presence, adequate, modified; (4) Presence, adequate, not applied; (5) Presence, inadequate	No 8%
Task and procedure training	(1) Absence; (2) Presence, adequate; (3) Presence, adequate, modified; (4) Presence, adequate, not applied; (5) Presence, inadequate	No 19%
PPE training	(1) Absence; (2) Presence, adequate; (3) Presence, adequate, modified; (4) Presence, adequate, not applied; (5) Presence, inadequate	No 2%
Confined space entry and procedures	(1) Absence; (2) Presence, adequate; (3) Presence, adequate, modified; (4) Presence, adequate, not applied; (5) Presence, inadequate	No 10%
Entry permit system	(1) Absence; (2) Presence, adequate; (3) Presence, adequate, modified; (4) Presence, adequate, not applied; (5) Presence, inadequate	No 6%
Lock out - tag out procedures	(1) Absence; (2) Presence, adequate; (3) Presence, adequate, modified; (4) Presence, adequate, not applied; (5) Presence, inadequate	No 19%
Warning signs	(1) Absence; (2) Presence, adequate; (3) Presence, adequate, modified; (4) Presence, adequate, not applied; (5) Presence, inadequate	No 4%
Atmosphere monitoring	(1) Absence; (2) Presence, adequate; (3) Presence, adequate, modified; (4) Presence, adequate, not applied; (5) Presence, inadequate	No 7%
Communication equipment and procedures	(1) Absence; (2) Presence, adequate; (3) Presence, adequate, modified; (4) Presence, adequate, not applied; (5) Presence, inadequate	No 4%
Attendant	(1) Absence; (2) Presence, adequate; (3) Presence, adequate, modified; (4) Presence, adequate, not applied; (5) Presence, inadequate	No 13%
Emergency procedures	(1) Absence; (2) Presence, adequate; (3) Presence, adequate, modified; (4) Presence, adequate, not applied; (5) Presence, inadequate	No 4%
Fall protection	(1) Absence; (2) Presence, adequate; (3) Presence, adequate, modified; (4) Presence, adequate, not applied; (5) Presence, inadequate	No 18%
Respiratory protection	(1) Absence; (2) Presence, adequate; (3) Presence, adequate, modified; (4) Presence, adequate, not applied; (5) Presence, inadequate	No 14%
Protective clothing	(1) Absence; (2) Presence, adequate; (3) Presence, adequate, modified; (4) Presence, adequate, not applied; (5) Presence, inadequate	No 4%
Other personal controls	(1) Absence; (2) Presence, adequate; (3) Presence, adequate, modified; (4) Presence, adequate, not applied; (5) Presence, inadequate	No 5%
Rescue equipment	(1) Absence; (2) Presence, adequate; (3) Presence, adequate, modified; (4) Presence, adequate, not applied; (5) Presence, inadequate	No 5%

Table A2

List, description, and classification of typical safety measures and barriers in confined spaces.

Safety measure/barrier	Description	Hierarchy of control	Safety function
Devices minimising the need for entry	Installation of devices to minimise the need for workers to climb or enter a confined space	Elimination	Avoid the entry
Use of less hazardous substances	Replacement of hazardous substances with less hazardous ones and adaptation to technical progress	Substitution	Avoid hazards
Use of less hazardous machinery or equipment	Replacement of hazardous equipment with less hazardous ones and adaptation to technical progress	Substitution	Avoid hazards
Ventilation systems	Installation of systems for natural and/or forced ventilation to guarantee an acceptable air exchange rate and avoid the accumulation of hazardous substances	Engineering controls	Prevent hazardous atmospheres
Purging of the atmosphere	Purging, inerting, drainage, flushing, or cleaning the space	Engineering controls	Prevent hazardous atmospheres
Equipment or position guarding	Guardrail, safety grid, electro-sensitive protective equipment, or other means for avoiding contacts between workers and equipment, and guarding working positions	Engineering controls	Prevent risks and exposure to hazards
Covers and barriers to avoid entry or fall	Cover and/or physical barriers to prevent unauthorised persons from entering or inadvertently falling into a confined space	Engineering controls	Prevent risks and exposure to hazards
Lighting systems	Placement of systems to assure a safe performance of work activities and safe entry and exit means, and to avoid ignition sources in flammable or explosive atmospheres	Engineering controls	Prevent risks and exposures to hazards
Isolation and containment systems	Isolation of the workers from hazards and implementation of containment systems to minimise leaks and hazardous substances near workers	Engineering controls	Prevent risks and exposures to hazards
Interlock systems	Systems for preventing system operation and start-up in unsafe or undesired conditions	Engineering controls	Prevent risks and exposures to hazards
Testing equipment	Equipment for monitoring the space and atmosphere to determine if acceptable conditions are present prior to and during entry into space	Engineering controls	Prevent hazardous atmospheres
Equipment maintenance	Maintenance schedule for equipment and machinery to address any failures before they occur	Administrative controls	Prevent risks and exposures to hazards
Confined space training	Training on confined space characteristics, hazards, risks, and procedures	Administrative controls	Prevent human errors
Task and procedure training	Training on the correct performance of tasks and work activities	Administrative controls	Prevent human errors
PPE training	Training on the correct use and maintenance of fall and respiratory protection equipment	Administrative controls	Prevent human errors
Confined space entry and procedures	Confined space entry, work procedures and programmes	Administrative controls	Prevent human errors
Entry permit system	Definition of an entry permit system or permit-required confined space identification and programme for managing authorised entry	Administrative controls	Prevent risks and exposures to hazards
Lock out - tag out procedures	Adoption of procedures and devices to control the operation, energisation, and undesired energy release from equipment and machinery	Administrative controls	Prevent risks and exposures to hazards
Warning signs	Signs summarising hazards, risks, and required procedures for safely enter the confined space	Administrative controls	Prevent human errors
Atmosphere monitoring	Periodical testing of the conditions inside the space and the atmosphere composition (i.e. oxygen, combustible gases and vapours, toxic gases and vapours)	Administrative controls	Detect the atmosphere
Communication equipment and procedures	Equipment and procedures to assure constant communication among workers inside and outside	Administrative controls	Control risks and exposures to hazards
Attendant	Outside standby person or helper monitoring entrants and maintaining contact with them	Administrative controls	Control risks and exposures to hazards
Emergency procedures	Rescue and emergency procedures, including training	Administrative controls	Limit exposures and consequences of undesired events
Fall protection	Harness, lifelines, safety belt, lanyard, access system	Personal Protective Equipment	Limit exposures and consequences of undesired events
Respiratory protection	Respirators, breathing apparatus, mask	Personal Protective Equipment	Limit exposures and consequences of undesired events
Protective clothing	Suit, waders	Personal Protective Equipment	Limit exposures and consequences of undesired events
Other personal controls	Safety glasses, welding hood, gloves, shoes, earmuffs, earplugs, other hearing protectors, hard hat	Personal Protective Equipment	Limit exposures and consequences of undesired events
Rescue equipment	Retrieval system, emergency retrieval harness	Personal Protective Equipment	Limit exposures and consequences of undesired events

References

- Botti, L., Mora, C., Ferrari, E., 2022. Design of a digital tool for the identification of confined spaces. *J. Loss Prev. Process Ind.* 76, 104731.
- Campari, A., Nakhal Akel, A.J., Ustolin, F., Alvaro, A., Ledda, A., Agnello, P., Moretto, P., Patriarca, R., Paltrinieri, N., 2023. Lessons learned from HIAD 2.0: Inspection and maintenance to avoid hydrogen-induced material failures. *Comput. Chem. Eng.* 173, 108199.
- Chowdhury, G.G., 2003. Natural language processing. *Annu. Rev. Inf. Sci. Technol.* 37 (1), 51–89.
- Delvosalle, C., Fiévez, C., Pipart, A., 2004. ARAMIS D1C - APPENDIX 08 Checklists of safety functions and barriers. Mons, Belgium.
- Demšar, J., Curk, T., Erjavec, A., Gorup, Č., Hočevar, T., Milutinovič, M., Možina, M., Polajnar, M., Toplak, M., Starič, A., Stajdohar, M., Umek, L., Žagar, L., Zbontar, J., Žitnik, M., Zupan, B., 2013. Orange.: Data Min. Toolbox Python. *J. Mach. Learn. Res.* 14, 2349–2353.

- ISO (International Organization for Standardization) ISO 31000:2018. Risk Management - Guidelines 2018a ISO, Geneva, Switzerland.
- ISO (International Organization for Standardization), IEC (International Electrotechnical Commission), 2014. ISO/IEC Guide 51:2014. Safety Aspects - Guidelines for Their Inclusion in Standards. ISO, IEC, Geneva, Switzerland.
- ISO (International Organization for Standardization) ISO 45001:2018. Occupational Health and Safety Management Systems - Requirements with Guidance for Use 2018b ISO, Geneva, Switzerland.
- European Commission, 2022. Commission Delegated Regulation (EU) 2022/692 of 16 February 2022 amending, for the purposes of its adaptation to technical and scientific progress, Regulation (EC) No 1272/2008 of the European Parliament and of the Council on classification, labelling and packaging of substances and mixtures. Brussels, Belgium.
- Gonzalez-Cortes, A., Burlet-Vienney, D., Chinniah, Y., Ben Mosbah, A., Bahloul, A., Ouellet, C., 2022. Inherently safer design (ISD) solutions in confined spaces: Experts' practical feedback in Quebec, Canada. *Process Saf. Environ. Prot.* 157, 375–389.
- HSE (Health and Safety Executive), 2013. Confined spaces. A brief guide to working safely. HSE, Suffolk, UK.
- Arunraj, N.S., Maiti, J., 2009. A methodology for overall consequence modeling in chemical industry. *J. Hazard. Mater.* 169, 556–574.
- Botti, L., Duraccio, V., Gnani, M.G., Mora, C., 2018. An integrated holistic approach to health and safety in confined spaces. *J. Loss Prev. Process Ind.* 55, 25–35.
- Breiman, L., 2001. Random Forests. *Mach. Learn.* 45, 5–32.
- Burlet-Vienney, D., Chinniah, Y., Bahloul, A., Roberge, B., 2015a. Design and application of a 5 step risk assessment tool for confined space entries. *Saf. Sci.* 80, 144–155.
- Burlet-Vienney, D., Chinniah, Y., Bahloul, A., Roberge, B., 2015b. Occupational safety during interventions in confined spaces. *Saf. Sci.* 79, 19–28.
- Chiu, C.-C., Chang, Y.-M., Wan, T.-J., 2020. Characteristic Analysis of Occupational Confined Space Accidents in Taiwan and Its Prevention Strategy. *Int. J. Environ. Res. Public Health* 17, 1752.
- ANSI (American National Standards Institute), ASSP (American Society of Safety Professionals), 2022. *ANSI/ASSP Z117.1-2022. Safety Requirements for Entering Confined Spaces*. ASSP, Park Ridge, IL.
- DPR (Decreto del Presidente della Repubblica) n. 177, 2011. Regolamento recante norme per la qualificazione delle imprese e dei lavoratori autonomi operanti in ambienti sospetti di inquinamento o confinati, a norma dell'articolo 6, comma 8, lettera g), del Decreto Legislativo 9 aprile 2008, n. 81 (in Italian).
- European Parliament and Council, 2008. Regulation (EC) of the European Parliament and of the Council of 16 December 2008 on classification, labelling and packaging of substances and mixtures, amending and repealing Directives 67/548/EEC and 1999/45/EC, and amending Regulation (EC) No 1907/2006. Strasbourg, France.
- Goasdoué, V., Nugier, S., Duquenois, D., Labois, B., 2007. An evaluation framework for data quality tools. In: *Proceedings of the 12th International Conference on Information Quality (ICIQ)*, Cambridge, MA, 280–294.
- Hoxmeier, J.A., 1998. Typology of database quality factors. *Softw. Qual. J.* 7, 179–193.
- ISO (International Organization for Standardization), IEC (International Electrotechnical Commission), 2022. ISO/IEC 23053:2022. Framework for Artificial Intelligence (AI) Systems Using Machine Learning (ML). ISO, Geneva, Switzerland.
- Jain, A.K., Murty, M.N., Flynn, P.J., 1999. Data clustering: A review. *ACM Comput. Surv.* 31 (3), 264–323.
- Kang, K., Ryu, H., 2019. Predicting types of occupational accidents at construction sites in Korea using random forest model. *Saf. Sci.* 120, 226–236.
- Khurana, D., Koli, A., Khatter, K., Singh, S., 2023. Natural language processing: state of the art, current trends and challenges. *Multimed. Tools Appl.* 82, 3713–3744.
- Kotsiantis, S.B., 2007. Supervised machine learning: a review of classification techniques. *Informatica* 31 (3), 249–268.
- Kotsiantis, S.B., Zaharakis, I.D., Pintelas, P.E., 2006. Machine learning: a review of classification and combining techniques. *Artif. Intell. Rev.* 26, 159–190.
- Li, X., Zhu, R., Ye, H., Jiang, C., Benslimane, A., 2021. MetaInjury: meta-learning framework for reusing the risk knowledge of different construction accidents. *Saf. Sci.* 140, 105315.
- Lindberg, A.-K., Hansson, S.O., Rollenhagen, C., 2010. Learning from accidents - what more do we need to know? *Saf. Sci.* 48, 714–721.
- Liu, G., Boyd, M., Yu, M., Halim, S.Z., Qudus, N., 2021. Identifying causality and contributory factors of pipeline incidents by employing natural language processing and text mining techniques. *Process Saf. Environ. Prot.* 152, 37–46.
- Maalouf, M., 2011. Logistic regression in data analysis: an overview. *Int. J. Data Anal. Tech. Strateg.* 3 (3), 281–299.
- McManus, N., 1998. Safety and health in confined spaces. CRC, Boca Raton, FL.
- McManus, N., 2010. Advancing the paradigm: confined spaces and the uncharacterized workspace. NorthWest Occupational Health and Safety, British Columbia, Canada.
- Naghavi Konjin, Z., Mortazavi, S.B., Asilian, M.H., Hajizadeh, E., 2019. Exploring the contributory factors of confined space accidents using accident investigation reports and semistructured interviews. *Saf. Health Work* 10, 305–313.
- Nakhal, A.J., Patriarca, R., Di Gravio, G., Antonioni, G., Paltrinieri, N., 2021. Investigating occupational and operational industrial safety data through Business Intelligence and Machine Learning. *J. Loss Prev. Process Ind.* 73, 104608.
- Ngah, H., Mohd Hairon, S., Hamzah, N.A., Noordin, S., Shafei, M.N., 2022. Development and validation of knowledge, attitude, and practice questionnaire: toward safe working in confined spaces. *Int. J. Environ. Res. Public Health* 19, 1242.
- NIOSH (National Institute for Occupational Safety and Health), 1979. *Criteria for a Recommended Standard... Working in Confined Space*. DHEW (NIOSH) Publication No. 80-106. US Department of Health, Education, and Welfare. Public Health Service. Center for Disease Control NIOSH, Cincinnati, OH.
- OSHA (Occupational Safety and Health Administration), 2011. Standard No. 29 CFR 1910.146. Permit-required confined spaces. OSHA, Washington, DC.
- Paltrinieri, N., 2022. Chapter Thirteen - Safety assessment of complex socio-technical systems. In: Khan, F., Pasman, H., Yang, M. (Eds.), *Methods to Assess and Manage Process Safety in Digitalized Process System*, 6. Elsevier, Cambridge, pp. 461–477.
- Paltrinieri, N., Comfort, L., Reniers, G., 2019. Learning about risk: machine learning for risk assessment. *Saf. Sci.* 118, 475–486.
- Pasman, H.J., 2009. Learning from the past and knowledge management: are we making progress? *J. Loss Prev. Process Ind.* 22, 672–679.
- Perlich, C., Provost, F., Simonoff, J.S., 2003. Tree Induction vs. Logistic Regression: A Learning-Curve Analysis. In: *Journal of Machine Learning Research*, 4, pp. 211–255.
- Rekus, J.F., 1994. *Complete Confined Spaces Handbook*. CRC, Boca Raton, FL.
- ISO (International Organization for Standardization), 2000. *ISO 11226. Ergonomics - Evaluation of Static Working Postures*. ISO, Geneva, Switzerland.
- SA (Standards Australia), 2009. *AS 2865-2009. Confined spaces*. SA, Sidney, Australia.
- Sammur, C., Webb, G.I., 2017. *Encyclopedia of Machine Learning and Data Mining*, Second edition. Springer, New York.
- Sarkar, S., Maiti, J., 2020. Machine learning in occupational accident analysis: a review using science mapping approach with citation network analysis. *Saf. Sci.* 131, 104900.
- Sarkar, S., Pramanik, A., Maiti, J., Reniers, G., 2020. Predicting and analyzing injury severity: a machine learning-based approach using class-imbalanced proactive and reactive data. *Saf. Sci.* 125, 104616.
- Secretary of State, Department of the Environment, Transport and the Regions, 1997. 1997 No. 1713. *Health and Safety. The Confined Spaces Regulations 1997*. United Kingdom.
- Selman, J., Spickett, J., Jansz, J., Mullins, B., 2018. An investigation into the rate and mechanism of incident of work-related confined space fatalities. *Saf. Sci.* 109, 333–343.
- Selman, J., Spickett, J., Jansz, J., Mullins, B., 2019. Confined space rescue: a proposed procedure to reduce the risks. *Saf. Sci.* 113, 78–90.
- Sen, P.C., Hajra, M., Ghosh, M., 2020. Supervised classification algorithms in machine learning: a survey and review. In: Mandal, J., Bhattacharya, D. (Eds.), *Emerging Technology in Modelling and Graphics, Advances in Intelligent Systems and Computing*, 937. Springer, Singapore, pp. 99–111.
- Simone, F., Ansaldi, S.M., Agnello, P., Patriarca, R., 2023. Industrial safety management in the digital era: constructing a knowledge graph from near misses. *Comput. Ind.* 146, 103849.
- Sklet, S., 2006. Safety barriers: Definition, classification, and performance. *J. Loss Prev. Process Ind.* 19, 494–506.
- Stefana, E., Marciano, F., Cocca, P., Alberti, M., 2018. Confined space risk management in steel industry: towards the adoption of Industry 4.0 technologies. In: *Proceedings of the XXIII Summer School "Francesco Turco"*, 94–100.
- Stefana, E., Marciano, F., Cocca, P., Rossi, D., Tomasoni, G., 2021a. Oxygen deficiency hazard in confined spaces in the steel industry: assessment through predictive models. *Int. J. Occup. Saf. Ergon.* 27 (4), 990–1004.
- Stefana, E., Marciano, F., Drolet, D., Armstrong, T.W., 2021b. A traditional near field-far field approach-based model and a spreadsheet workbook to manage Oxygen Deficiency Hazard. *Process Saf. Environ. Prot.* 149, 537–556.
- Stefana, E., Ghidoni, D., Fanizza, F., Marciano, F., Cocca, P., Paltrinieri, N., 2022a. Towards Machine Learning Application for Safety in Confined Spaces: Creating an Incident Database. In: Leva, M.C., Patelli, E., Podofilini, L., Wilson, S. (Eds.), *Proceedings of the 32nd European Safety and Reliability Conference*. Research Publishing, Singapore, pp. 2686–2693.
- Stefana, E., Ustolin, F., Paltrinieri, N., 2022b. IMPROSafety: a risk-based framework to integrate occupational and process safety. *J. Loss Prev. Process Ind.* 75, 104698.
- Stefana, E., Patriarca, R., Droguett, E.L., Paltrinieri, N., 2023. Editorial on meta-learning peculiarities and approaches for facing safety issues and enhancing risk management process. *Saf. Sci.* 159, 106040.
- Suruda, A.J., Pettit, T.A., Noonan, G.P., Ronk, R.M., 1994. Deadly rescue: the confined space hazard. *J. Hazard. Mater.* 36, 45–53.
- Tamascelli, N., Paltrinieri, N., Cozzani, V., 2020. Predicting chattering alarms: a machine learning approach. *Comput. Chem. Eng.* 143, 107122.
- Tamascelli, N., Paltrinieri, N., Cozzani, V., 2023. Learning from major accidents: a meta-learning perspective. *Saf. Sci.* 158, 105984.
- US Department of Labor - Bureau of Labor Statistics, 2012. *Occupational Injury and Illness Classification Manual*. Version 2.01.
- United Nations, Department of Economic and Social Affairs, 2008. *International Standard Industrial Classification of All Economic Activities*. In: Revision 4. United Nations, New York, NY.
- Wang, B., Zhao, J., 2022. Automatic frequency estimation of contributory factors for confined space accidents. *Process Saf. Environ. Prot.* 157, 193–207.
- Witten, I.H., Frank, E., Hall, M.A., Pal, C.J., 2017. In: Kaufmann, Morgan (Ed.), *Data Mining. Practical Machine Learning Tools and Techniques*, Fourth edition. Elsevier, Cambridge.
- Wu, X., Kumar, V., Ross Quinlan, J., Ghosh, J., Yang, Q., Motoda, H., McLachlan, G.J., Ng, A., Liu, B., Yu, P.S., Zhou, Z.-H., Steinbach, M., Hand, D.J., Steinberg, D., 2008. Top 10 algorithms in data mining. *Knowl. Inf. Syst.* 14, 1–37.
- Xia, J., Liu, Y., Zhao, D., Tian, Y., Li, J., Zhong, Y., Roy, N., 2021. Human factors analysis of China's confined space operation accidents from 2008 to 2018. *J. Loss Prev. Process Ind.* 71, 104480.
- Yan, L., Yantek, D.S., DeGennaro, C.R., Fernando, R.D., 2023. Mathematical Modeling for Carbon Dioxide Level Within Confined Spaces. *ASCE-ASME J. Risk Uncertain. Eng. Syst., Part B: Mech. Eng.* 9, 024501.
- Ying, X., 2019. An overview of overfitting and its solutions. *J. Phys.: Conf. Ser.* 1168 (2), 022022.
- Zhang, Y., Yang, Q., 2022. A survey on multi-task learning. *IEEE Trans. Knowl. Data Eng.* 34 (12), 5586–5609.

Web References

- CAMEO Chemicals. *Database of Hazardous Materials*. <<https://cameochemicals.noaa.gov/>> (Accessed 04.23).
- CSB (US Chemical Safety Board). *Investigations*. <<https://www.csb.gov/investigations/>> (Accessed 04.23).
- INAIL (Italian National Institute for Insurance against Accidents at Work). *Infortuni gravi*. <<https://www.inail.it/sol-informo/home.do?tipoEvento=0>> (Accessed 04.23) (in Italian).
- INAIL (Italian National Institute for Insurance against Accidents at Work). *Infortuni mortali*. <<https://www.inail.it/sol-informo/home.do?tipoEvento=1>> (Accessed 04.23) (in Italian).
- NIOSH (US National Institute for Occupational Safety & Health). *Fatality Assessment and Control Evaluation (FACE) Program. State FACE Reports: Confined Space*. <<https://www.cdc.gov/NIOSH-FACE/Default.cshhtml?Category=0004&Category2=ALL&Submit=Submit>> (Accessed 04.23).
- NIOSH (US National Institute for Occupational Safety & Health). *Fatality Assessment and Control Evaluation (FACE) Program. NIOSH FACE Reports: Confined Space*. <https://www.cdc.gov/NIOSH-FACE/Default.cshhtml?state=ALL&Incident_Year=ALL&Category2=0004&Submit=Submit> (Accessed 04.23).
- Orange. *Download Orange*. <<https://orangedatamining.com/>> (Accessed 04.23).
- OSHA (US Occupational Safety and Health Administration). *Severe Injury Reports*. <<https://www.osha.gov/severeinjury/>> (Accessed 04.23).
- OSHA (US Occupational Safety and Health Administration). *Fatality and Catastrophe Investigation Summaries*. <<https://www.osha.gov/ords/imis/accidentsearch.html>> (Accessed 04.23).